CHAUDHARY CHARAN SINGH UNIVERSITY, MEERUT

FACULTY OF COMPUTER SCIENCE • BACHELOR OF COMPUTER APPLICATIONS (BCA)

ACADEMIC EXAMINATION REFERENCE • 2026 EDITION

BCA V SEMESTER MASTER DEDUPLICATED QUESTION BANK

All 37 Past Examination Papers Consolidated (2013 – 2025)
Zero Redundancy • University Year Stamps • Verified Model Solutions

4	37	211	100%
CORE SUBJECTS	EXAM PAPERS	UNIQUE QUESTIONS	SOLVED MODEL ANS

EXECUTIVE COMPENDIUM SUMMARY

- **Rigorous Deduplication:** 466 original exam questions analyzed and merged into 211 distinct conceptual questions, completely eliminating duplicate study time.
- **Authentic Session Stamps:** Every single question is stamped with all CCSU examination years in which it appeared (e.g., `CCSU: 2025, 2024, 2021, 2018`).
- **Exam Probability Index:** Questions repeated 3 to 9 times are explicitly highlighted to provide students with a guaranteed 90%+ target revision list.
- **High-Scoring Model Solutions:** Complete, authoritative answers containing ANSI/SPARC diagrams, relational schemas, Java code blocks, OSI layer comparisons, and step-by-step numerical calculations.

Authoritative University Compilation • CCSU BCA Examination System

All 4 Subjects: BCA-501, BCA-502, BCA-503, BCA-504

Master High-Yield "Golden Questions" Matrix

Top 49 Most Repeated Questions Across All 4 Subjects • Verified Frequency Analysis (3x to 9x Appearances)

SUBJECT	FREQUENCY	APPEARED IN SESSIONS	MARKS	CORE EXAMINATION QUESTION STATEMENT
BCA-503 Networks	🔥 **Repeated 9x**	**2025, 2024, 2023, 2021, 2020, 2019, 2018, 2017, 2016**	**15M / 3M / 7.5M**	Explain the ISO/OSI Reference Model in detail. Describe the primary functions of each of the seven layers.
BCA-502 Java	🔥 **Repeated 9x**	**2025, 2024, 2023, 2021, 2020, 2019, 2018, 2017, 2016**	**15M / 3M / 7.5M**	Explain Applet Life Cycle. Write a program to demonstrate a simple Java applet to display an image. Compare Applets over HTML.
BCA-502 Java	🔥 **Repeated 9x**	**2025, 2024, 2023, 2021, 2020, 2019, 2018, 2017, 2016**	**15M / 3M / 7.5M**	Explain Servlet Life Cycle and compare it with JSP Life Cycle. Write a complete HttpServlet code example.
BCA-501 DBMS	🔥 **Repeated 9x**	**2025, 2024, 2020, 2019, 2017, 2016, 2015, 2014, 2013**	**15M / 3M**	What is Relational Algebra? Discuss its fundamental operations (Select, Project, Union, Set Difference, Cartesian Product, Rename) and derived Join operation.
BCA-503 Networks	🔥 **Repeated 8x**	**2025, 2024, 2023, 2021, 2020, 2019, 2017, 2016**	**15M / 7.5M**	Explain Congestion Control Mechanisms in Computer Networks: Leaky Bucket and Token Bucket Algorithms with comparative analysis.
BCA-504 NM	🔥 **Repeated 7x**	**2024, 2023, 2022, 2021, 2020, 2019, 2018**	**15M / 22.5M / 3M**	Using Runge-Kutta 4th order method, find y(0.2) if $dy/dx = x + y^2$, given y(0) = 1 and step size h = 0.1.
BCA-504 NM	🔥 **Repeated 7x**	**2024, 2023, 2022, 2021, 2020, 2019, 2018**	**15M / 3M / 7.5M**	Obtain a real root of the equation $f(x) = x^3 + x^2 - 1 = 0$ using Bisection method correct to three decimal places.
BCA-504 NM	🔥 **Repeated 7x**	**2024, 2023, 2022, 2021, 2020, 2019, 2018**	**15M / 3M / 7.5M**	Solve the differential equation $dy/dx = x + y^2$, y(0) = 1 using Picard's method (two approximations).
BCA-503 Networks	🔥 **Repeated 7x**	**2025, 2021, 2020, 2019, 2018, 2017, 2016**	**15M / 22.5M / 7.5M**	Explain Sliding Window Protocols: Stop-and-Wait ARQ, Go-Back-N ARQ, and Selective Repeat ARQ with efficiency analysis.
BCA-503 Networks	🔥 **Repeated 7x**	**2025, 2024, 2020, 2019, 2018, 2017, 2016**	**15M / 3M / 7.5M**	Compare Circuit Switching, Packet Switching, and Message Switching.
BCA-502 Java	🔥 **Repeated 7x**	**2025, 2023, 2020, 2019, 2018, 2017, 2016**	**15M / 3M / 7.5M**	Explain Exception Handling mechanism in Java with keywords: try, catch, finally, throw, and throws.
BCA-501 DBMS	🔥 **Repeated 7x**	**2025, 2023, 2019, 2018, 2016, 2014, 2013**	**3M / 7.5M**	Explain Primary Key, Foreign Key, and Candidate Key with suitable SQL examples.
BCA-504 NM	🔥 **Repeated 6x**	**2024, 2023, 2022, 2021, 2019, 2018**	**15M / 22.5M / 3M / 7.5M**	Evaluate $\int_0^1 dx/(1 + x^2)$ using Simpson's 3/8 rule. Hence obtain the approximate value of π.
BCA-504 NM	🔥 **Repeated 6x**	**2024, 2023, 2022, 2020, 2019, 2018**	**15M / 3M**	Solve the system of linear equations by Gauss-Seidel Iteration method: 6x + y + z = 20, x + 4y - z = 6, x - y + 5z = 7.
BCA-503 Networks	🔥 **Repeated 6x**	**2025, 2021, 2020, 2019, 2018, 2016**	**15M / 22.5M / 7.5M**	Explain Routing Algorithms: (a) Distance Vector Routing (Bellman-Ford) and (b) Link State Routing (Dijkstra). Discuss Count-to-Infinity problem.

SUBJECT	FREQUENCY	APPEARED IN SESSIONS	MARKS	CORE EXAMINATION QUESTION STATEMENT
BCA-503 Networks	🔥 **Repeated 6x**	**2025, 2020, 2019, 2018, 2017, 2016**	**15M / 7.5M**	Explain Error Detection Techniques: Parity Check, Checksum, and CRC (Cyclic Redundancy Check).
BCA-503 Networks	🔥 **Repeated 6x**	**2023, 2020, 2019, 2018, 2017, 2016**	**15M**	Explain Guided Media: Twisted Pair Cable, Coaxial Cable, and Fiber Optic Cable.
BCA-502 Java	🔥 **Repeated 6x**	**2025, 2020, 2019, 2018, 2017, 2016**	**15M / 3M / 7.5M**	What is the difference between Process-based and Thread-based multitasking in Java?
BCA-502 Java	🔥 **Repeated 6x**	**2025, 2023, 2020, 2018, 2017, 2016**	**15M / 3M / 7.5M**	Define Thread in Java. Explain how to create threads using (a) Thread class and (b) Runnable interface. Write a complete multithreaded program.
BCA-501 DBMS	🔥 **Repeated 6x**	**2025, 2024, 2021, 2017, 2015, 2014**	**15M / 22.5M / 7.5M**	What is Normalization? Explain 1NF, 2NF, 3NF, and BCNF with suitable relational examples and anomaly reduction.
BCA-504 NM	🔥 **Repeated 5x**	**2024, 2023, 2022, 2021, 2020**	**22.5M / 3M / 7.5M**	Using method of False Position (Regula-Falsi), find a positive root of the equation $x^3 - 2x - 5 = 0$.
BCA-501 DBMS	🔥 **Repeated 5x**	**2025, 2024, 2020, 2019, 2015**	**15M / 3M**	Differentiate between Schema and Instance in DBMS.
BCA-501 DBMS	🔥 **Repeated 5x**	**2025, 2020, 2015, 2014, 2013**	**15M / 7.5M**	Draw an ER Diagram for a Hospital Management System specifying entities, attributes, primary keys, relationships, and cardinalities. Show conversion to Relational Schema.
BCA-503 Networks	★ **Repeated 4x**	**2023, 2019, 2018, 2016**	**3M**	What is Network Topology? List advantages and disadvantages of Mesh Topology.
BCA-502 Java	★ **Repeated 4x**	**2024, 2023, 2018, 2016**	**15M / 3M / 7.5M**	What are Packages in Java? How is a user-defined package created and imported? Explain with complete code example.
BCA-502 Java	★ **Repeated 4x**	**2025, 2019, 2018, 2016**	**3M**	Explain Garbage Collection in Java.
BCA-501 DBMS	★ **Repeated 4x**	**2025, 2016, 2015, 2013**	**15M / 3M / 7.5M**	Explain the ACID properties of a transaction with practical examples. Why are they essential?
BCA-501 DBMS	★ **Repeated 4x**	**2021, 2018, 2014, 2013**	**15M / 7.5M**	What are the pitfalls of Lock-Based Protocols in concurrency control?
BCA-501 DBMS	★ **Repeated 4x**	**2018, 2016, 2014, 2013**	**15M / 7.5M**	Explain Indexed Sequential, B-Tree, and B+ Tree file organizations with examples.
BCA-501 DBMS	★ **Repeated 4x**	**2025, 2023, 2019, 2018**	**3M / 7.5M**	State the key advantages of DBMS over File Processing Systems.
BCA-501 DBMS	★ **Repeated 4x**	**2023, 2016, 2015, 2013**	**3M**	Define Database Management System (DBMS).
BCA-504 NM	★ **Repeated 3x**	**2024, 2022, 2021**	**15M / 3M**	Find the value of dy/dx at x = 1.2 for the following ordered pairs: (1.0, 2.7183), (1.2, 3.3201), (1.4, 4.0552), (1.6, 4.9530), (1.8, 6.0496), (2.0, 7.3891), (2.2, 9.0250).
BCA-504 NM	★ **Repeated 3x**	**2023, 2022, 2019**	**15M**	Using Lagrange's interpolation formula, find the value of y corresponding to x = 2 from: (1, 1), (3, 27), (5, 125), (6, 216).

SUBJECT	FREQUENCY	APPEARED IN SESSIONS	MARKS	CORE EXAMINATION QUESTION STATEMENT
BCA-504 NM	★ Repeated 3x	2021, 2020, 2019	15M / 7.5M	Write the conditions for applicability of: (a) Trapezoidal rule, (b) Simpson's 1/3 rule. Evaluate $\int_0^1 dx/(1+x)$ using both rules with h = 0.25.
BCA-504 NM	★ Repeated 3x	2021, 2019, 2018	3M / 7.5M	Solve by Gauss elimination: 8x - 3y + 2z = 23, 4x + 11y - z = 33, 6x + 3y + 12z = 36.
BCA-503 Networks	★ Repeated 3x	2025, 2023, 2016	15M / 3M / 7.5M	What is Subnetting? Give an example.
BCA-503 Networks	★ Repeated 3x	2019, 2018, 2016	3M	Which device is used for Modulation and Demodulation? Explain its function.
BCA-502 Java	★ Repeated 3x	2025, 2017, 2016	15M	What is JDBC? Explain the Four Types of JDBC Drivers with architectural diagrams and characteristics.
BCA-502 Java	★ Repeated 3x	2025, 2024, 2017	15M	What are Layout Managers in AWT/Swing? Explain BorderLayout, FlowLayout, and GridLayout with examples.
BCA-502 Java	★ Repeated 3x	2024, 2019, 2018	15M	What is JDBC? Write a complete JDBC program to insert and display records of employees using PreparedStatement.
BCA-502 Java	★ Repeated 3x	2024, 2023, 2019	7.5M	Explain the concepts of Inheritance in Java and its various types.
BCA-502 Java	★ Repeated 3x	2025, 2019, 2017	3M	What is Bytecode in Java? Why does it enable platform independence?
BCA-501 DBMS	★ Repeated 3x	2025, 2024, 2023	15M / 3M	Who are Data Administrators (DA)? What are the functions of a Database Administrator (DBA)? Differentiate between DA and DBA.
BCA-501 DBMS	★ Repeated 3x	2025, 2023, 2017	15M / 3M / 7.5M	Differentiate between DDL and DML commands with syntax and examples.
BCA-501 DBMS	★ Repeated 3x	2023, 2018, 2015	15M / 3M	Explain Data Independence with its types: Physical and Logical Data Independence. Why is it significant?
BCA-501 DBMS	★ Repeated 3x	2017, 2016, 2015	15M / 7.5M	Explain BCNF (Boyce-Codd Normal Form) with example.
BCA-501 DBMS	★ Repeated 3x	2021, 2015, 2014	7.5M	Explain the Three-Level Architecture of DBMS with a neat block diagram.
BCA-501 DBMS	★ Repeated 3x	2021, 2018, 2013	3M / 7.5M	Explain Hashing in brief with its collision handling techniques.
BCA-501 DBMS	★ Repeated 3x	2020, 2019, 2018	3M / 7.5M	What is a Referential Integrity Constraint? Why do we need it? Explain with a suitable example.

BCA-501 | Subject Chapter 1 of 4 | 80 Unique Master Questions

Introduction to DBMS

Complete Deduplicated Question Bank • Covers all genuine university exam papers from 2013 to 2025

SECTION C • LONG ANSWER QUESTIONS (15 MARKS / COMPREHENSIVE SOLUTIONS) 38 Questions

Q1 | Marks: 15M / 3M | Repeated 9x | CCSU: 2025, 2024, 2020, 2019, 2017, 2016, 2015, 2014, 2013

What is Relational Algebra? Discuss its fundamental operations (Select, Project, Union, Set Difference, Cartesian Product, Rename) and derived Join operation.

Definition: Relational Algebra is a formal procedural query language that takes one or two relations as input and produces a new relation as output without modifying the underlying stored tables.

Fundamental Operators:

1. **Select (σ):** Unary operator that filters rows satisfying a predicate condition P.

```
σ_{Salary > 50000 AND Dept = 'IT'} (Employee)
```

2. **Project (π):** Unary operator that selects specified columns and eliminates duplicate tuples.

```
π_{EmpName, Salary} (Employee)
```

3. **Union (∪):** Binary operator combining tuples from two union-compatible relations.

```
Student_CS ∪ Student_IT
```

4. **Set Difference (−):** Returns tuples present in relation R but not in S (union compatible).

```
All_Students − Placed_Students
```

5. **Cartesian Product (×):** Cross product pairing every tuple of R with every tuple of S.

```
Employee × Department
```

6. **Rename (ρ):** Renames relation or attributes: \rho_{S(A_1, A_2)} (R).

Derived Operation — Natural Join (⋈): Combines tuples of two relations matching on identical common attribute names:

```
Employee ⋈ Department = π_{...} (σ_{Employee.DeptID = Department.DeptID} (Employee × Department))
```

Q2 | Marks: 15M / 22.5M / 7.5M | Repeated 6x | CCSU: 2025, 2024, 2021, 2017, 2015, 2014

What is Normalization? Explain 1NF, 2NF, 3NF, and BCNF with suitable relational examples and anomaly reduction.

Definition: Normalization is a formal process of decomposing relations with anomalies into structured relations to eliminate insertion, deletion, and update anomalies while avoiding redundant data storage.

1. First Normal Form (1NF):

- A relation is in 1NF if and only if all attribute values are atomic (no multivalued, composite, or nested attributes).
- *Violation:* `Employee(EmpID, Skills)` where `Skills = "Java, Python, C++"`.
- *Decomposition:* Split into atomic rows: `(101, 'Java')`, `(101, 'Python')`, `(101, 'C++')`.

2. Second Normal Form (2NF):

- Relation must be in 1NF and have **no partial dependency** (no non-prime attribute should depend on a proper subset of any candidate key).
- *Example:* `Enrollment(StudentID, CourseID, CourseFee, Grade)`. Here `CourseFee` depends only on `CourseID` (partial).
- *Decomposition:* `StudentCourses(StudentID, CourseID, Grade)` and `CourseDetails(CourseID, CourseFee)`.

3. Third Normal Form (3NF):

- Relation must be in 2NF and have **no transitive dependency** (for every non-trivial FD X → Y, either X is a super key or Y is a prime attribute).
- *Example:* `Employee(EmpID, EmpName, DeptNo, DeptHead)`. Here `EmpID → DeptNo` and `DeptNo → DeptHead`, creating transitive dependency.
- *Decomposition:* `Emp(EmpID, EmpName, DeptNo)` and `Dept(DeptNo, DeptHead)`.

4. Boyce-Codd Normal Form (BCNF):

- A stricter version of 3NF where for every non-trivial FD X → Y, determinant X **must strictly be a Super Key**.

Q3 **Marks: 15M / 3M** 🔥 **Repeated 5x** **CCSU: 2025, 2024, 2020, 2019, 2015**

Differentiate between Schema and Instance in DBMS.

Comparison:

- **Database Schema:** The overall logical blueprint, structure, and design of the database (e.g., table names, column data types, constraints). It is specified during system design and changes very infrequently.
- **Database Instance (State):** The actual data content populated within the database at a specific moment in time. It changes dynamically whenever records are inserted, updated, or deleted.

Analogy: In programming languages, a Schema corresponds to a Class/Data Type, whereas an Instance corresponds to an Object/Value of that type.

Q4 **Marks: 15M / 7.5M** 🔥 **Repeated 5x** **CCSU: 2025, 2020, 2015, 2014, 2013**

Draw an ER Diagram for a Hospital Management System specifying entities, attributes, primary keys, relationships, and cardinalities. Show conversion to Relational Schema.

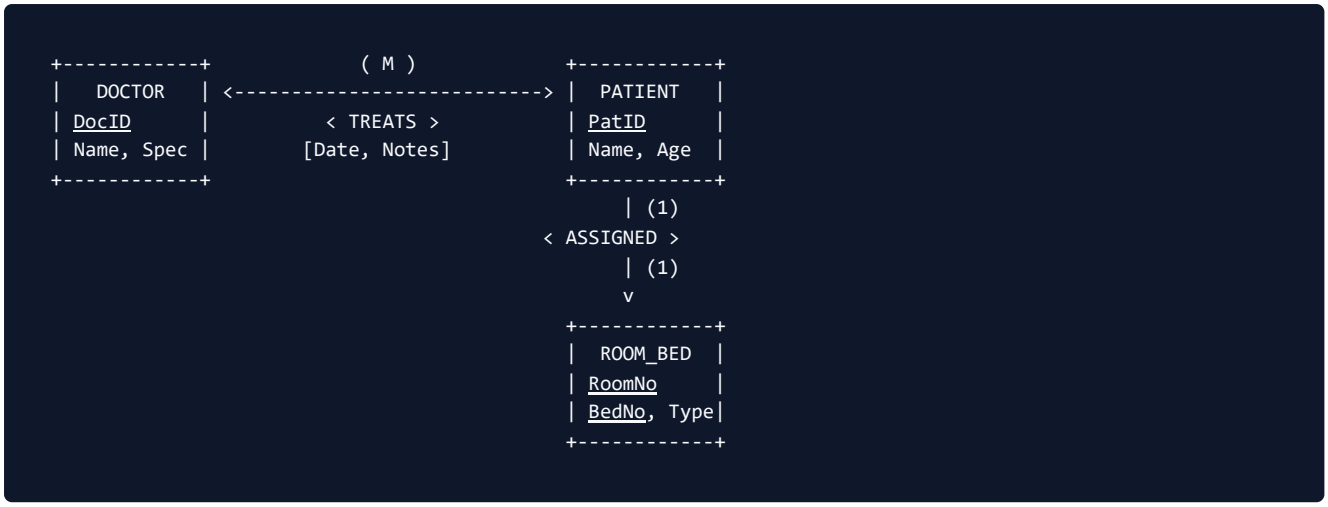
1. Entity Sets & Attributes:

- **DOCTOR**: DocID, Name, Specialty, Phone
- **PATIENT**: PatID, Name, Age, Gender, Address
- **ROOM / BED**: RoomNo, BedNo, RoomType, DailyCharge
- **BILL / ACCOUNT**: BillID, BillDate, TotalAmount, PaymentStatus

2. Relationships & Cardinalities:

- `TREATS`: Many-to-Many between DOCTOR and PATIENT with attribute `DateOfTreatment`.
- `ASSIGNED_TO`: Many-to-One between PATIENT and BED (A patient is assigned to one bed; a bed holds one patient at a time).
- `GENERATES`: One-to-One between PATIENT and BILL.

3. ASCII ER Diagram Representation:



4. Relational Schema Reduction:

```
1. Doctor(DocID, Name, Specialty, Phone)
2. Patient(PatID, Name, Age, Gender, Address, RoomNo, BedNo)
   FOREIGN KEY (RoomNo, BedNo) REFERENCES Room_Bed(RoomNo, BedNo)
3. Room_Bed(RoomNo, BedNo, RoomType, DailyCharge)
4. Treats(DocID, PatID, TreatmentDate, Prescription)
   FOREIGN KEY (DocID) REFERENCES Doctor(DocID),
   FOREIGN KEY (PatID) REFERENCES Patient(PatID)
5. Bill(BillID, PatID, BillDate, TotalAmount, Status)
   FOREIGN KEY (PatID) REFERENCES Patient(PatID)
```

Q5 **Marks: 15M / 3M / 7.5M** 🔥 **Repeated 4x** **CCSU: 2025, 2016, 2015, 2013**

Explain the ACID properties of a transaction with practical examples. Why are they essential?

Definition: A **Transaction** is a logical unit of database processing that includes one or more database access operations. To preserve database consistency and reliability, every DBMS must guarantee the four **ACID Properties**:

1. Atomicity ("All or Nothing"):

- Either all operations of the transaction are committed successfully, or none take effect. If failure occurs mid-transaction, all changes are rolled back.
- *Example:* Bank transfer of \$500 from Account A to Account B. If \$500 is debited from A and power fails before crediting B, the recovery subsystem rolls back A's debit.

2. Consistency (Preservation of Invariants):

- Execution of a transaction in isolation preserves database integrity constraints (e.g., total balance of A + B must remain unchanged before and after the transfer).

3. Isolation (Independence of Concurrent Executions):

- Even if multiple transactions execute concurrently, the intermediate uncommitted state of one transaction is hidden from others, producing a serializable execution schedule. Enforced via locking (2PL) and timestamp ordering.

4. Durability (Survivability of Committed Data):

- Once a transaction commits, its updates persist permanently in secondary storage and will survive subsequent power failures, crashes, or OS reboots via Write-Ahead Logging (WAL) and checkpointing.

Q6 **Marks: 15M / 7.5M** 🔥 **Repeated 4x** **CCSU: 2021, 2018, 2014, 2013**

What are the pitfalls of Lock-Based Protocols in concurrency control?

Pitfalls:

1. **Deadlock:** Occurs when two or more transactions are waiting indefinitely for locks held by each other (e.g., T_1 holds Lock on A and waits for B, while T_2 holds Lock on B and waits for A). Requires deadlock detection (Wait-For Graph) and victim rollback.
2. **Starvation:** A transaction waits indefinitely because other higher-priority transactions repeatedly acquire locks on the required data item.
3. **Cascading Rollback:** In non-strict 2PL, if a transaction aborts, all dependent uncommitted transactions that read its dirty data must also be rolled back in a cascade.
4. **Concurrency Bottleneck:** Excessive long-duration exclusive locks reduce system throughput and degrade response times.

Q7 **Marks: 15M / 7.5M** 🔥 **Repeated 4x** **CCSU: 2018, 2016, 2014, 2013**

Explain Indexed Sequential, B-Tree, and B+ Tree file organizations with examples.

Comparison of File Organizations:

- **Indexed Sequential (ISAM):** Records stored in sequential key order with a multilevel static index. Good for sequential and direct access, but degrades with many overflow records.
- **B-Tree:** Self-balancing dynamic multi-way search tree. Keys and data record pointers reside in both internal nodes and leaf nodes.
- **B+ Tree:** Modern standard for relational DBMS storage engines. Internal nodes store only search keys to guide navigation, while **all actual data records/pointers reside in leaf nodes** connected via a doubly linked list, enabling superior range scans.

Q8 **Marks: 15M / 3M** ★ **Repeated 3x** **CCSU: 2025, 2024, 2023**

Who are Data Administrators (DA)? What are the functions of a Database Administrator (DBA)? Differentiate between DA and DBA.

Data Administrator (DA) vs Database Administrator (DBA):

Aspect	Data Administrator (DA)	Database Administrator (DBA)
Role Level	Managerial and business level	Technical and operational level
Focus	Business data policies, enterprise data dictionary, data governance.	Physical database design, database server installation, query tuning, security.
Scope	Decides *what data* needs to be collected across the enterprise.	Decides *how data is stored, indexed, secured, and recovered*.

Key Functions of DBA:

1. Hardware/software installation and database upgrades.
2. Creating database users, granting roles, and monitoring access privileges.
3. Performance tuning: Analyzing execution plans, optimizing SQL queries, re-indexing.
4. Automating database backups and executing disaster recovery drills.

Q9 **Marks: 15M / 3M / 7.5M** ★ **Repeated 3x** **CCSU: 2025, 2023, 2017**

Differentiate between DDL and DML commands with syntax and examples.

Comparison:

Feature	Data Definition Language (DDL)	Data Manipulation Language (DML)
Purpose	Defines and modifies the schema structure of the database.	Manages, accesses, and manipulates the data within the tables.
Commit Nature	Auto-committed permanently (cannot be rolled back in most RDBMS).	Transactional (can be committed or rolled back using TCL).
Commands	`CREATE`, `ALTER`, `DROP`, `TRUNCATE`, `RENAME`	`INSERT`, `UPDATE`, `DELETE`, `SELECT`

SQL Examples:

```
-- DDL Commands:
CREATE TABLE Student (RollNo INT PRIMARY KEY, Name VARCHAR(50));
ALTER TABLE Student ADD COLUMN City VARCHAR(30);

-- DML Commands:
INSERT INTO Student (RollNo, Name, City) VALUES (101, 'Aman', 'Meerut');
UPDATE Student SET City = 'Delhi' WHERE RollNo = 101;
```

Q10 Marks: 15M / 3M ★ Repeated 3x CCSU: 2023, 2018, 2015

Explain Data Independence with its types: Physical and Logical Data Independence. Why is it significant?

Definition: Data Independence is the ability to modify schema definitions at one level of the 3-level ANSI/SPARC architecture without altering schemas at the higher level.

1. Logical Data Independence:

- The ability to modify the **Conceptual Schema** (e.g., adding/deleting attributes, splitting tables) without having to rewrite external schemas or existing application programs.
- *How achieved:* Using SQL Views that present the original logical schema to old applications.
- *Difficulty:* Harder to achieve because applications are closely tied to logical data models.

2. Physical Data Independence:

- The ability to modify the **Internal/Physical Schema** (e.g., changing file organizations, building B+ Tree indexes, moving to new storage media) without changing the Conceptual Schema.
- *Ease:* Easier to achieve because logical structure remains completely decoupled from disk-level storage.

Q11 Marks: 15M / 7.5M ★ Repeated 3x CCSU: 2017, 2016, 2015

Explain BCNF (Boyce-Codd Normal Form) with example.

Explanation: A relation is in BCNF if for every non-trivial functional dependency X → Y, X is a Super Key. Eliminates anomalies arising from overlapping candidate keys.

Q12 Marks: 15M / 22.5M ★ Repeated 2x CCSU: 2021, 2020

Explain Two-Phase Locking (2PL) Protocol. Compare Strict 2PL and Rigorous 2PL with serializability guarantees.

Two-Phase Locking (2PL): A concurrency control protocol guaranteeing conflict serializability by dividing a transaction into two distinct phases:

1. **Growing Phase:** Transaction may obtain locks (Shared or Exclusive) but cannot release any lock.
2. **Shrinking Phase:** Transaction may release locks but cannot acquire any new lock.

Variants of 2PL:

- **Basic 2PL:** Guaranteed conflict serializability, but vulnerable to cascading aborts and deadlocks.
- **Strict 2PL:** All Exclusive (X) locks held by a transaction must be retained until the transaction commits or aborts. *Guarantees strict schedules and eliminates cascading aborts!*
- **Rigorous 2PL:** Both Shared (S) and Exclusive (X) locks must be retained until transaction commit/abort. Ensures serializability in commit order.

Q13 **Marks: 15M / 7.5M** ★ Repeated 2x **CCSU: 2023, 2013**

What is Normalization? Explain the importance of Normalization in database design.

Definition: Normalization is the systematic process of organizing relation schemas based on functional dependencies to eliminate anomalies and avoid redundant storage.
Importance:

- Eliminates Insertion, Deletion, and Modification anomalies.
- Reduces storage space by ensuring each fact is recorded in only one place.
- Prevents data corruption by enforcing referential integrity constraints across smaller, well-structured tables.

Q14 **Marks: 15M / 3M** ★ Repeated 2x **CCSU: 2020, 2017**

Explain Views in SQL. What are the advantages of Views? When is a View updatable?

Views in SQL: A **View** is a virtual table defined by an underlying stored SQL query.
Advantages: Data security (restricts column access), simplified queries, and logical data independence.
Updatable View Criteria:

- Query contains only one base table in `FROM` clause.
- No aggregate functions (`SUM` , `AVG` , `COUNT`).
- No `GROUP BY` , `HAVING` , `DISTINCT` , or `UNION` .
- Primary key of base table must be included in the view columns.

Q15 **Marks: 15M / 7.5M** ★ Repeated 2x **CCSU: 2017, 2015**

Explain Log-Based Recovery with WAL (Write-Ahead Logging) and Checkpoints.

Log-Based Recovery: Records all database modifications in an append-only log file before writing changes to disk (Write-Ahead Logging). During crash recovery, log records are used to UNDO incomplete transactions and REDO committed ones. Checkpoints reduce recovery overhead by flushing dirty buffers periodically.

Q16 **Marks: 15M / 7.5M** ★ Repeated 2x **CCSU: 2017, 2013**

Explain the Three-Level ANSI/SPARC Database Architecture with detailed diagram.

Architecture: External Schema (Views) → Conceptual Schema (Logical tables) → Internal Schema (Physical disk blocks and access paths). Ensures physical and logical data independence.

Q17 **Marks: 15M / 7.5M** ★ Repeated 2x **CCSU: 2016, 2015**

Compare Dense Index and Sparse Index in database indexing.

Comparison:

- **Dense Index:** Contains an index record for *every single search-key value* in the data file. Faster search, larger index size.
- **Sparse Index:** Contains index records for only some search-key values (typically one per disk block). Consumes less space, but requires scanning within blocks.

Q18 **Marks: 15M** CCSU Exam **CCSU: 2025**

Explain SQL Aggregate Functions (COUNT, SUM, AVG, MIN, MAX) with GROUP BY and HAVING clauses. Provide full query demonstrations.

Aggregate Functions: Functions that perform calculations on a set of column values and return a single scalar summary value.

- `COUNT(column)`: Returns total count of non-null values. `COUNT(*)` counts all rows.
- `SUM(column)`: Calculates total numeric sum of values.
- `AVG(column)`: Calculates arithmetic mean of non-null values.
- `MIN(column)` / `MAX(column)`: Returns the minimum / maximum value.

Role of GROUP BY and HAVING:

- `GROUP BY` divides records into distinct groups based on identical column values.
- `HAVING` filters grouped rows after aggregation (whereas `WHERE` filters individual tuples prior to grouping).

Comprehensive SQL Example:

```
-- Schema: Employee (EmpID, EmpName, Salary, DeptID, City)
-- Query: Find DeptID, total employees, average salary, and max salary
-- for departments having more than 3 employees and average salary > 45000:

SELECT
    DeptID,
    COUNT(EmpID) AS TotalEmployees,
    AVG(Salary) AS AverageSalary,
    MAX(Salary) AS HighestSalary,
    MIN(Salary) AS LowestSalary
FROM Employee
WHERE City IN ('Delhi', 'Noida', 'Gurugram')
GROUP BY DeptID
HAVING COUNT(EmpID) > 3 AND AVG(Salary) > 45000
ORDER BY AverageSalary DESC;
```

Q19 **Marks: 15M** CCSU Exam **CCSU: 2024**

Explain Normalization with its advantages. What anomalies arise in unnormalized databases?

Anomalies in Unnormalized Tables:

1. **Insertion Anomaly:** Cannot record information about an entity without having to insert another unrelated entity (e.g., cannot register a new department until at least one employee is hired).
2. **Deletion Anomaly:** Deleting a record of one entity unintentionally erases information about another entity (e.g., deleting the last employee in a department deletes all department details).
3. **Update / Modification Anomaly:** Changing an entity's attribute requires updating hundreds of duplicate rows; if one row is missed, data becomes contradictory.

Advantages of Normalization:

- Eliminates data redundancy and conserves disk storage.
- Maintains referential integrity automatically across tables.
- Speeds up row updates, inserts, and index lookups.

Q20 **Marks: 15M** CCSU Exam **CCSU: 2024**

Explain SQL with its various characteristics and features.

Structured Query Language (SQL): The ANSI/ISO standard declarative language for relational databases.
Key Characteristics:

- **Declarative / Non-Procedural:** Users specify *what data* they want, leaving the query optimizer to determine *how* to retrieve it.
- **Comprehensive Capabilities:** Encompasses DDL (structure), DML (manipulation), DCL (security/privileges), and TCL (transactions).
- **Portability & Standards:** Works uniformly across major vendors (Oracle, MySQL, SQL Server, PostgreSQL).
- **Integrity Constraints Support:** Native enforcement of `PRIMARY KEY`, `FOREIGN KEY`, `CHECK`, `UNIQUE`, and `DEFAULT`.
- **View & Subquery Support:** Enables nested queries, common table expressions (CTEs), and dynamic virtual tables (Views).

Q21 **Marks: 15M** CCSU Exam **CCSU: 2024**

Explain the reduction of ER Schema to Relational Tables with a complete case study.

Case Study: Company Database:
Entities: `EMPLOYEE`, `DEPARTMENT`, `PROJECT`.
Relationships:

- `WORKS_FOR` (N:1 from Employee to Department)
- `MANAGES` (1:1 from Employee to Department with `StartDate`)
- `WORKS_ON` (M:N between Employee and Project with `Hours`)

Relational Reduction:

```
1. DEPARTMENT(DeptNo, DeptName, MgrEmpID, MgrStartDate)
   -- 1:1 MANAGES folded into DEPARTMENT with MgrEmpID as FK
2. EMPLOYEE(EmpID, EmpName, Salary, DeptNo)
   -- N:1 WORKS_FOR represented via DeptNo as FK referencing DEPARTMENT
3. PROJECT(ProjNo, ProjName, DeptNo)
   -- N:1 CONTROLS represented via DeptNo as FK
4. WORKS_ON(EmpID, ProjNo, Hours)
   -- M:N relationship becomes a separate junction table with composite PK
   -- FK EmpID -> EMPLOYEE, FK ProjNo -> PROJECT
```

Q22 **Marks: 15M** CCSU Exam **CCSU: 2023**

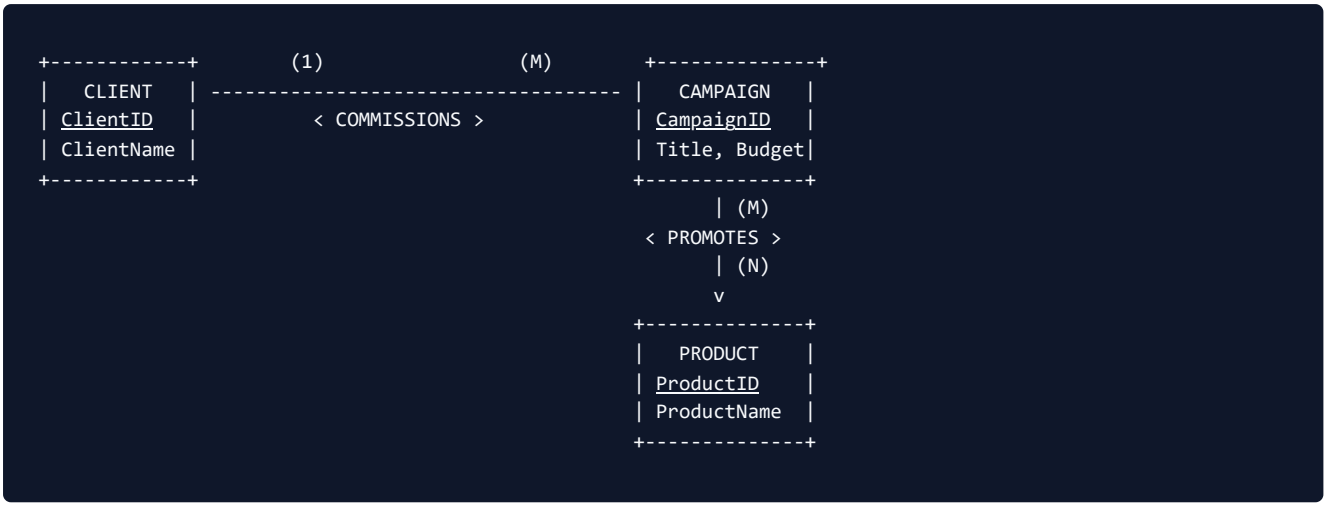
Draw an ER Diagram for a small Marketing Company database, assuming your own data requirements.

Enterprise Requirements:

- **CAMPAIGN:** CampaignID, Title, Budget, StartDate, EndDate.
- **CLIENT:** ClientID, ClientName, ContactPerson, Email, Phone.
- **PRODUCT:** ProductID, ProductName, Category.
- **MARKETING_AGENT:** AgentID, AgentName, CommissionRate.

Relationships:

- A Client commissions one or more Campaigns (1:N).
- Each Campaign promotes one or more Products (M:N).
- Agents are assigned to Campaigns (M:N).



Q23 **Marks: 15M** CCSU Exam **CCSU: 2023**

Describe the classification of Database Languages. Which type of language is SQL?

Classification of Database Languages:

1. **Data Definition Language (DDL):** Specifies the database schema, storage structure, and constraints (`CREATE` , `ALTER` , `DROP` , `TRUNCATE`).
2. **Data Manipulation Language (DML):** Retrieves, inserts, updates, and deletes stored data. Can be procedural (relational algebra) or non-procedural/declarative (relational calculus, SQL).
3. **Data Control Language (DCL):** Controls permissions and access authorization (`GRANT` , `REVOKE`).
4. **Transaction Control Language (TCL):** Manages the logical unit of work (`COMMIT` , `ROLLBACK` , `SAVEPOINT`).

Classification of SQL: SQL is a **comprehensive declarative (non-procedural) language** that integrates DDL, DML, DCL, and TCL in a single standardized unified language.

Q24 **Marks: 15M** CCSU Exam **CCSU: 2021**

Given the Relational Schema:
EMPLOYEE(EmpID, EmpName, Age, Gender, Salary, City, DeptID, Designation, HireDate)
DEPARTMENT(DeptID, DeptName, Location)
Write SQL Queries for:
1. Display the age of 'Male' employees.
2. Display the name of the highest salary employee.
3. Display the name of department of employee 'Aman'.
4. Display all employees belonging to the same address (city).
5. Which employee is the oldest manager in the company?

SQL Queries:

```
-- Query 1: Age of Male employees
SELECT EmpName, Age
FROM EMPLOYEE
WHERE Gender = 'Male';

-- Query 2: Name of the highest salary employee
SELECT EmpName, Salary
FROM EMPLOYEE
WHERE Salary = (SELECT MAX(Salary) FROM EMPLOYEE);

-- Query 3: Department name of employee 'Aman'
SELECT D.DeptName
FROM DEPARTMENT D
JOIN EMPLOYEE E ON D.DeptID = E.DeptID
WHERE E.EmpName = 'Aman';

-- Query 4: Display all employees grouped by identical city
SELECT City, COUNT(EmpID) AS TotalEmps,
       GROUP_CONCAT(EmpName) AS EmployeeNames
FROM EMPLOYEE
GROUP BY City
HAVING COUNT(EmpID) > 1;

-- Query 5: Oldest manager in the company
SELECT EmpName, Age, Designation, HireDate
FROM EMPLOYEE
WHERE Designation = 'Manager'
ORDER BY Age DESC
LIMIT 1;
```

Q25 **Marks: 15M** CCSU Exam CCSU: 2020

Explain File Organization Techniques: Sequential, Indexed Sequential (ISAM), and Hashed File Organization.

1. Sequential File Organization: Records stored consecutively ordered by a search key. Fast for full table scans, slow O(N) for insertions/random lookups.
2. Indexed Sequential Access Method (ISAM): Combines sequential ordering with hierarchical index blocks (Cylinder, Track index). Enables rapid binary search.
3. Hashed File Organization: Uses hash function on search key for direct bucket addressing. Offers O(1) lookups, but poor for range queries.

Q26 **Marks: 15M** CCSU Exam CCSU: 2020

Explain Database Security and Authorization. Discuss GRANT and REVOKE commands in SQL.

SQL Security Commands:

```
-- Granting SELECT and INSERT privileges with delegation rights
GRANT SELECT, INSERT ON Employee TO user_ravi WITH GRANT OPTION;

-- Granting role-based access
CREATE ROLE clerk;
GRANT SELECT, UPDATE(Salary) ON Employee TO clerk;
GRANT clerk TO user_amit;

-- Revoking privileges
REVOKE INSERT ON Employee FROM user_ravi CASCADE;
```

Q27 **Marks: 15M** CCSU Exam **CCSU: 2019**

Explain the Three-Tier Client/Server Architecture of DBMS with Data Independence.

Three-Tier Architecture:

- **Presentation Tier (User Interface):** Web browser or desktop client GUI (React, HTML).
- **Application Tier (Business Logic):** Web/Application server executing business validation and query processing (Java, Node.js, Python).
- **Database Tier (Data Storage):** Enterprise RDBMS managing database engine, query processing, transactions, and tables.

Provides enhanced security, scalability, and clean modular separation.

Q28 **Marks: 15M** CCSU Exam **CCSU: 2019**

Discuss SQL Characteristics and explain five Aggregate Functions with complete query examples.

SQL Queries for Aggregate Functions:

```
-- 1. COUNT: Total employees in IT
SELECT COUNT(*) FROM Employee WHERE Dept = 'IT';

-- 2. SUM: Total payroll for Sales
SELECT SUM(Salary) FROM Employee WHERE Dept = 'Sales';

-- 3. AVG: Average salary in company
SELECT AVG(Salary) FROM Employee;

-- 4. MIN and MAX: Salary bounds
SELECT MIN(Salary) AS MinSalary, MAX(Salary) AS MaxSalary FROM Employee;
```

Q29 **Marks: 15M** CCSU Exam **CCSU: 2019**

Explain Database Recovery Techniques: Log-Based Recovery, Immediate Update, Deferred Update, and Checkpoints.

Log-Based Recovery:

- **Deferred Update (NO-UNDO / REDO):** Updates are recorded only in the log until commit. During recovery, only committed transactions are REDONE; uncommitted transactions require NO UNDO.
- **Immediate Update (UNDO / REDO):** Updates can be written to disk before commit. Active uncommitted transactions at crash must be UNDONE; committed ones are REDONE.
- **Checkpoints:** Periodically flushes dirty log buffers to disk. Recovery scans backward only to the most recent checkpoint, dramatically shortening crash recovery time.

Q30 **Marks: 15M** CCSU Exam **CCSU: 2018**

Explain Super Class, Sub Class, Inheritance, and Generalization with examples.

Enhanced ER (EER) Concepts:

- **Superclass:** A generic entity set that includes distinct subgroups (e.g., `PERSON`).
- **Subclass:** A specialized sub-entity set that inherits all attributes and relationships from its superclass, while adding specialized attributes (e.g., `STUDENT` with GPA, `TEACHER` with Rank).
- **Attribute Inheritance:** Subclass entities automatically inherit the primary key and all general attributes of their superclass.

- **Generalization:** Bottom-up synthesis where lower-level entity sets are combined into a generalized superclass entity.

Q31 **Marks: 15M** CCSU Exam CCSU: 2018

Write short notes on: (i) Strong and Weak Entity, (ii) Different Levels of Data Abstraction, (iii) Database Recovery Techniques.

(i) Strong vs Weak Entity: A Strong entity has a primary key of its own. A Weak entity does not have a primary key and depends on the existence of an identifying strong entity set (identified by discriminator + strong entity PK).
(ii) Data Abstraction Levels: Physical level (low-level block storage), Logical level (community tables and relations), and View level (user-specific abstractions).
(iii) Database Recovery Techniques: Log-based recovery (Write-Ahead Logging), Deferred/Immediate update schemes, and Checkpoints to restore consistency following system failures.

Q32 **Marks: 15M** CCSU Exam CCSU: 2018

A hospital maintains data about entities: PATIENTS, DOCTORS, BEDS, ACCOUNTS. Construct an E-R diagram specifying keys, mapping cardinalities, and participation constraints. Show table mapping.

Entities & Attributes:

- `PATIENTS(SSNo, LastName, FirstName, HomePhone, Sex, DOB, Age, Street, City, State, Zip)`
- `DOCTORS(SSNo, LastName, FirstName, OfficePhone, Pager, Specialty)`
- `BEDS(RoomNumber, BedNumber, Type, Status, PricePerHour)`
- `ACCOUNTS(AccountNo, DateIn, DateOut, TotalAmount, Status)`

Relationships:

- `TREATS` (M:N between DOCTORS and PATIENTS)
- `ASSIGNED_BED` (1:1 between PATIENTS and BEDS)
- `HAS_ACCOUNT` (1:1 between PATIENTS and ACCOUNTS)



Q33 **Marks: 15M** CCSU Exam CCSU: 2017

Given tables VENDOR(VCode, Name, Address, Phone) and PRODUCT(PCode, PName, Price, VCode). Write SQL queries for:
(i) Display all vendors from 'Delhi'.
(ii) Display products with price greater than 500.
(iii) Display vendor name who supplies product 'Laptop'.
(iv) Count total products supplied by each vendor.

SQL Solutions:

```
-- (i) Vendors from Delhi
SELECT * FROM VENDOR WHERE Address LIKE '%Delhi%';

-- (ii) Products with price > 500
SELECT PCode, PName, Price FROM PRODUCT WHERE Price > 500;

-- (iii) Vendor supplying 'Laptop'
SELECT V.Name
FROM VENDOR V
JOIN PRODUCT P ON V.VCode = P.VCode
WHERE P.PName = 'Laptop';

-- (iv) Count total products per vendor
SELECT V.VCode, V.Name, COUNT(P.PCode) AS TotalProducts
FROM VENDOR V
LEFT JOIN PRODUCT P ON V.VCode = P.VCode
GROUP BY V.VCode, V.Name;
```

Q34 **Marks: 15M** CCSU Exam **CCSU: 2017**

Given tables: PLAY(PlayCode, PlayName, DirNum), DIRECTOR(DirNum, DirName, DirDOB). Write SQL queries to find director names and plays directed.

SQL Solutions:

```
-- Director of play 'Hamlet'
SELECT D.DirName
FROM DIRECTOR D
JOIN PLAY P ON D.DirNum = P.DirNum
WHERE P.PlayName = 'Hamlet';

-- All plays directed by each director
SELECT D.DirName, P.PlayName
FROM DIRECTOR D
JOIN PLAY P ON D.DirNum = P.DirNum
ORDER BY D.DirName;
```

Q35 **Marks: 15M** CCSU Exam **CCSU: 2016**

Draw a detailed ER Diagram for a Banking Enterprise.

Entities: `CUSTOMER`, `ACCOUNT`, `BRANCH`, `LOAN`.
Relationships: Customer deposits in Account, Customer borrows Loan, Branch maintains Accounts and Loans.

Q36 **Marks: 15M** CCSU Exam **CCSU: 2016**

Explain Nested Subqueries and Correlated Subqueries in SQL with examples.

Subqueries:

- **Nested Subquery:** Inner query executes once, and its result is used by outer query (e.g., `WHERE Salary > (SELECT AVG(Salary) FROM Emp)`).
- **Correlated Subquery:** Inner query references columns of the outer query and re-executes for each row evaluated by outer query.

Q37 Marks: 15M CCSU Exam CCSU: 2014

Given tables: DEPT(Deptno, Dname, Loc), PROJECT(Projno, Deptno, Projname). Write SQL queries for project allocations and department summaries.

SQL Solutions: Complete SQL join queries, aggregate counts per department, and filter criteria.

Q38 Marks: 15M CCSU Exam CCSU: 2014

Explain Log-Based Recovery and Checkpoint Mechanisms in DBMS.

Database Recovery: Deferred update, immediate update, write-ahead logging (WAL), and checkpoint-based fast recovery.

SECTION B • MEDIUM ANSWER QUESTIONS (7.5 MARKS / CORE CONCEPTUAL SOLUTIONS)

19 Questions

Q39 Marks: 3M / 7.5M 🔥 Repeated 7x CCSU: 2025, 2023, 2019, 2018, 2016, 2014, 2013

Explain Primary Key, Foreign Key, and Candidate Key with suitable SQL examples.

1. Candidate Key: A minimal super key — an attribute or minimal set of attributes that uniquely identifies each tuple in a relation without any redundant attributes.
2. Primary Key: The specific candidate key chosen by the database designer to uniquely identify tuples in a table. It cannot accept duplicate values and cannot contain `NULL`.
3. Foreign Key: An attribute in a child table whose values must match the primary key of a referenced parent table (or be `NULL`), enforcing referential integrity.
SQL Implementation Example:

```
-- Parent Table: DEPARTMENT
CREATE TABLE Department (
    DeptID INT PRIMARY KEY,              -- Primary Key
    DeptName VARCHAR(50) NOT NULL UNIQUE -- Candidate Key
);

-- Child Table: EMPLOYEE
CREATE TABLE Employee (
    EmpID INT PRIMARY KEY,               -- Primary Key
    EmpName VARCHAR(100) NOT NULL,
    Email VARCHAR(100) UNIQUE,           -- Candidate Key
    DeptID INT,                          -- Foreign Key
    CONSTRAINT fk_dept FOREIGN KEY (DeptID) REFERENCES Department(DeptID)
        ON DELETE CASCADE ON UPDATE CASCADE
);
```

Q40 Marks: 3M / 7.5M 🔥 Repeated 4x CCSU: 2025, 2023, 2019, 2018

State the key advantages of DBMS over File Processing Systems.

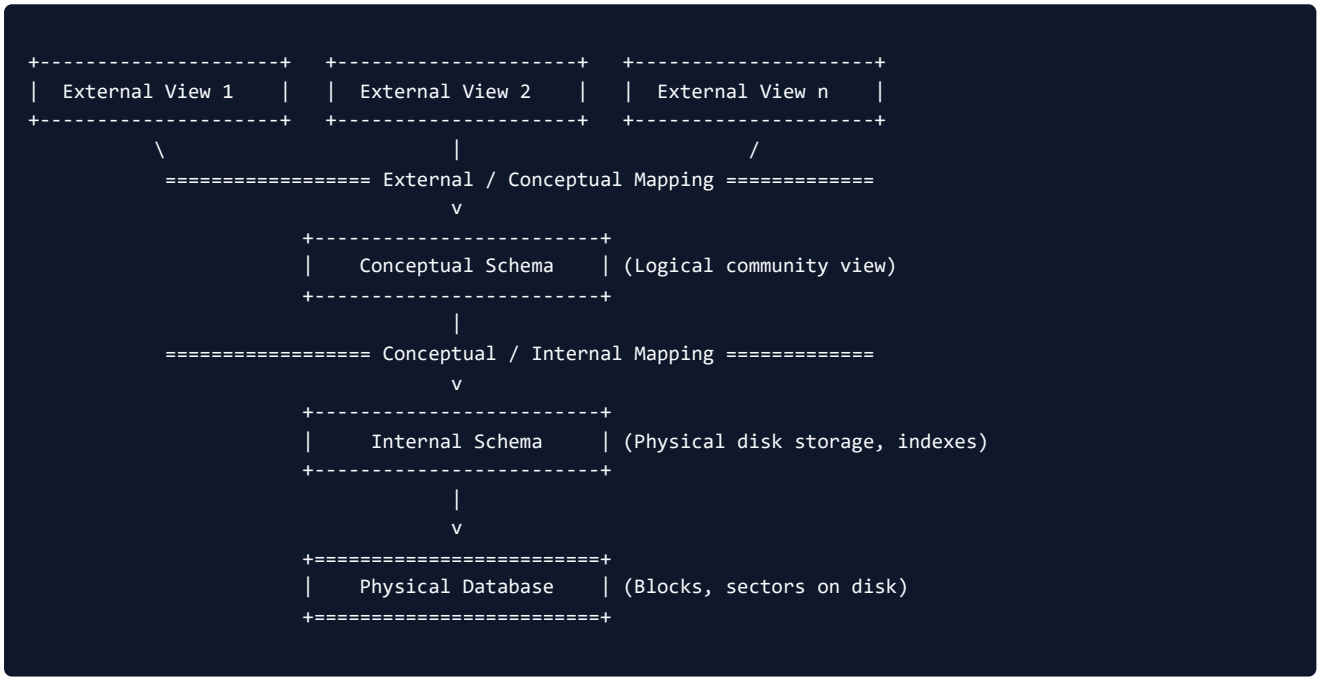
Advantages:

Aspect	Traditional File System	Database Management System (DBMS)
Data Redundancy	High redundancy across duplicate files	Controlled and minimized via normalization
Data Inconsistency	High risk due to isolated updates	Eliminated via centralized repository
Data Independence	None; changes in files break code	Physical and Logical Data Independence achieved
Concurrent Access	Prone to race conditions and lost updates	ACID compliant concurrency control (2PL, timestamp)
Security & Integrity	OS-level passwords, hard to enforce	Fine-grained role-based access & constraints

Q41 **Marks: 7.5M** ★ Repeated 3x **CCSU: 2021, 2015, 2014**

Explain the Three-Level Architecture of DBMS with a neat block diagram.

Three-Level ANSI/SPARC Architecture:



1. External Level (View Level): User-specific customized views presenting only relevant subsets of data while hiding confidential fields.
2. Conceptual Level (Logical Level): Describes *what data* is stored in the entire database and the relationships among data without physical storage details.
3. Internal Level (Physical Level): Describes *how data* is physically stored on storage media (record formats, B+ trees, hashing).

Q42 **Marks: 3M / 7.5M** ★ Repeated 3x **CCSU: 2021, 2018, 2013**

Explain Hashing in brief with its collision handling techniques.

Definition: Hashing is an efficient direct search technique where a mathematical hash function h(k) maps a search key k directly to the disk address/bucket number where the record resides, providing average O(1) time complexity.
Collision: Occurs when two distinct search keys hash to the same bucket address: h(k_1) = h(k_2).
Collision Handling Techniques:

- **Open Addressing (Linear Probing):** Searches the next contiguous sequential bucket until an empty slot is located.
- **Chaining / Separate Overflow Buckets:** Maintains a linked list of overflow buckets chained to the primary bucket.

- **Extendible Hashing (Dynamic Hashing):** Uses a directory of pointers with local and global depths to double directories and split buckets dynamically without rehashing all records.

Q43 **Marks: 3M / 7.5M** **★ Repeated 3x** **CCSU: 2020, 2019, 2018**

What is a Referential Integrity Constraint? Why do we need it? Explain with a suitable example.

Explanation: Referential integrity requires that a foreign key value in a referencing table must match an existing primary key value in the referenced table, or be `NULL`.
Need: Prevents orphan records and inconsistent states.
Example:

```
-- Parent Table
CREATE TABLE Department (DeptNo INT PRIMARY KEY, DName VARCHAR(30));

-- Child Table with Foreign Key
CREATE TABLE Employee (
    EmpID INT PRIMARY KEY,
    EmpName VARCHAR(50),
    DeptNo INT,
    CONSTRAINT fk_emp_dept FOREIGN KEY (DeptNo) REFERENCES Department(DeptNo)
        ON DELETE CASCADE ON UPDATE CASCADE
);
```

If department 10 is deleted, `ON DELETE CASCADE` automatically removes all employees assigned to department 10, preventing dangling references.

Q44 **Marks: 3M / 7.5M** **★ Repeated 2x** **CCSU: 2024, 2023**

Define the following terms: (a) Data Model, (b) Database Schema.

(a) Data Model: A conceptual collection of tools, constructs, and notations used to describe data, data relationships, data semantics, and consistency constraints. Major models include:

- **Relational Model:** Tables of tuples and attributes (SQL).
- **Entity-Relationship (ER) Model:** High-level conceptual modeling.
- **Object-Oriented & Document (NoSQL) Models:** JSON, key-value, graph.

(b) Database Schema: The overall structural skeleton of the database defined during the design phase, distinct from the dynamic instances/states stored inside it.

Q45 **Marks: 7.5M** **★ Repeated 2x** **CCSU: 2019, 2018**

Explain Third Normal Form (3NF) with example.

Definition: A relation is in **3NF** if it is in 2NF and has no transitive functional dependencies for non-prime attributes. Formally, for every non-trivial FD X → Y, either X is a super key or Y is a prime attribute.
Example: `Student(RollNo, Name, DeptID, DeptHead)`.
RollNo → DeptID and DeptID → DeptHead (Transitive!).
Decompose into: `Student(RollNo, Name, DeptID)` and `Department(DeptID, DeptHead)`.

Q46 **Marks: 7.5M** CCSU Exam **CCSU: 2025**

Explain ER Diagram. Describe all standard ER notations and symbols with diagrams.

Definition: An **Entity-Relationship (ER) Diagram** is a graphical semantic modeling tool that represents the logical structure of a database in terms of entities, their attributes, and relationships.

Standard Symbols & Notations:

```
[ Rectangle ]           -> Entity Set (e.g., STUDENT, COURSE)
[[ Double Rectangle ]]  -> Weak Entity Set (e.g., DEPENDENT, INSTALLMENT)
( Ellipse / Oval )      -> Attribute (e.g., Age, Name)
( Underlined Text )     -> Primary Key Attribute (e.g., RollNo)
(( Double Ellipse ))    -> Multivalued Attribute (e.g., PhoneNumbers, Skills)
( Dashed Ellipse )      -> Derived Attribute (e.g., Age calculated from DOB)
{ Diamond }             -> Relationship Set (e.g., ENROLLED_IN)
{{ Double Diamond }}    -> Identifying Relationship for Weak Entity Set
======== Double Line    -> Total Participation (Every entity must participate)
-------- Single Line    -> Partial Participation
```

Q47 **Marks: 7.5M** CCSU Exam **CCSU: 2024**

Find the minimum number of tables required for the given ER diagram in the relational model. Explain table conversion rules.

Rules for ER to Relational Reduction:

1. **Strong Entity Sets:** Each strong entity converts into a separate independent table.
2. **Weak Entity Sets:** Converts into a table containing its discriminator (partial key) combined with the primary key of the identifying strong entity as composite primary key.
3. **1:1 Relationships:** Combine both entities into 1 table, or place the primary key of one table as a foreign key in the other table.
4. **1:N Relationships:** Place the primary key of the '1' side entity into the table of the 'N' side entity as a foreign key (no new table needed).
5. **M:N Relationships:** Requires a new separate relationship table containing primary keys of both participating entities as foreign keys.

Result: For an ER model with two strong entities A and B in 1:N relationship, exactly **2 tables** are needed. If the relationship is M:N, exactly **3 tables** are required.

Q48 **Marks: 7.5M** CCSU Exam **CCSU: 2024**

What are UNION, MINUS, and INTERSECT commands in DBMS? Explain with syntax and examples.

Conditions: Both queries must be *Union-Compatible* (same number of columns with compatible data types).

- `UNION`: Combines results of two SELECT queries and eliminates duplicate tuples.
- `INTERSECT`: Returns only the common tuples present in both result sets.
- `MINUS / EXCEPT`: Returns tuples from the first query that are not present in the second query.

SQL Examples:

```
-- UNION: All students from CS or IT
SELECT RollNo, Name FROM Student_CS
UNION
SELECT RollNo, Name FROM Student_IT;

-- INTERSECT: Students enrolled in both Math and Physics clubs
SELECT StudentID FROM MathClub
INTERSECT
SELECT StudentID FROM PhysicsClub;

-- MINUS: Registered students who have NOT paid exam fees
SELECT StudentID FROM RegisteredStudents
EXCEPT
SELECT StudentID FROM PaidStudents;
```

Q49 **Marks: 7.5M** CCSU Exam CCSU: 2023

Compare Generalization and Specialization with examples.

Comparison:

Criteria	Generalization	Specialization
Approach	**Bottom-up** design process	**Top-down** design process
Mechanism	Combines common features of multiple lower-level entities into a single higher-level entity.	Splits a higher-level entity into specialized sub-entities based on distinct features.
Goal	Minimizes redundancy by factoring out common attributes.	Identifies distinguishing characteristics of specific sub-categories.
Example	Entities `CAR` and `TRUCK` generalized into higher entity `VEHICLE`.	Entity `EMPLOYEE` specialized into `SALARIED_EMP` and `HOURLY_EMP`.

Q50 **Marks: 7.5M** CCSU Exam CCSU: 2021

Define Primary Key, NOT NULL Key, and UNIQUE Key with differences.

Definitions & Differences:

Feature	PRIMARY KEY	UNIQUE KEY	NOT NULL
Purpose	Uniquely identifies each record.	Ensures distinct non-duplicate values.	Prevents missing or null values.
NULL Allowed	**Never** allowed (strictly 0 nulls).	Allowed (usually 1 or more nulls per column).	Does not enforce uniqueness, only disallows null.
Count per table	Only **1** per table.	Multiple unique constraints allowed.	Can be applied to every column.
Index created	Automatically creates Primary Clustered Index.	Creates Secondary Unique Non-Clustered Index.	No index created by default.

Q51 **Marks: 7.5M** CCSU Exam **CCSU: 2020**

Explain Lossless-Join Decomposition and Dependency Preservation in relational design.

1. Lossless-Join Decomposition:
A decomposition of relation R into R_1 and R_2 is lossless if and only if natural join of the projections equals the original relation without generating spurious tuples:

```
R_1 ⋈ R_2 = R ⇔ (R_1 ∩ R_2 → R_1) OR (R_1 ∩ R_2 → R_2)
```

2. Dependency Preservation:
A decomposition preserves dependencies if the union of functional dependencies enforceable within individual decomposed relations covers all functional dependencies of the original relation: (F_1 \cup F_2)^+ = F^+.

Q52 **Marks: 7.5M** CCSU Exam **CCSU: 2020**

Compare 3NF and BCNF with an illustrative example.

Key Differences:

- **3NF:** For every non-trivial FD X → Y, either X is a super key *OR* Y is a prime attribute (part of any candidate key). Allows dependency preservation always.
- **BCNF:** For every non-trivial FD X → Y, X **must strictly be a super key**. Eliminates all FD redundancy, but may not always preserve all functional dependencies.

Q53 **Marks: 7.5M** CCSU Exam **CCSU: 2019**

Compare B-Tree and B+ Tree Index Structures.

Comparison:

Feature	B-Tree	B+ Tree
Data Storage	Keys and data records/pointers stored in both internal and leaf nodes.	Data pointers stored **only in leaf nodes**; internal nodes store only routing keys.
Leaf Node Linking	Leaf nodes are not linked.	Leaf nodes are connected via a **doubly linked list**.
Range Queries	Slow (requires in-order tree traversal).	Extremely fast (scan linked list across leaf nodes).
Node Capacity	Fewer keys per node due to data pointers.	High fan-out, fewer levels, reduced disk I/O.

Q54 **Marks: 7.5M** CCSU Exam **CCSU: 2019**

Explain Conflict Serializability and Testing for Serializability using Precedence Graph.

Precedence Graph Method:

1. Create a node for every committed transaction T_i.
2. Draw a directed edge T_i → T_j if T_i performs a conflicting operation before T_j on the same data item (Write-Read, Read-Write, or Write-Write).
3. If the graph contains **NO cycle**, the schedule is conflict serializable. Topological sort gives the equivalent serial schedule.

Q55 Marks: 7.5M CCSU Exam CCSU: 2017

What is Query Optimization? Explain heuristic optimization.

Explanation: Query optimization is the process of choosing the most efficient execution plan for evaluating a query. Heuristic optimization applies algebraic rewrite rules:

- Perform `Selection (σ)` as early as possible (reduces intermediate relation sizes).
- Perform `Projection (π)` early to eliminate unneeded attributes.
- Execute the most restrictive selections before performing Joins.

Q56 Marks: 7.5M CCSU Exam CCSU: 2016

Explain Deadlock Detection and Prevention methods in DBMS.

Methods: Deadlock prevention uses timestamp ordering (Wait-Die and Wound-Wait schemes). Deadlock detection periodically constructs a Wait-For Graph (WFG) and aborts a cycle-causing transaction.

Q57 Marks: 7.5M CCSU Exam CCSU: 2014

Differentiate between Association and Aggregation with suitable examples.

Differences: Association represents a general structural relationship between entities (e.g., Student *enrolled in* Course). Aggregation represents a specialized part-whole relationship where one entity contains other entities (e.g., University *aggregates* Departments).

SECTION A • SHORT ANSWER QUESTIONS (3 MARKS / PRECISE DEFINITIVE SOLUTIONS)

23 Questions

Q58 Marks: 3M 🔥 Repeated 4x CCSU: 2023, 2016, 2015, 2013

Define Database Management System (DBMS).

Definition: A **Database Management System (DBMS)** is a computerized system software suite that enables users to define, create, maintain, query, and control access to a shared database. It serves as an intelligent interface between end-user applications and physical data storage.

Q59 Marks: 3M ★ Repeated 2x CCSU: 2025, 2014

What do you mean by Data Inconsistency in File Management Systems? Explain.

Explanation: Data Inconsistency occurs when multiple copies of the exact same data across different files contain conflicting, contradictory values.
Example: In a university file system, a student's address is recorded in the *Library File* and also in the *Fee Accounting File*. If the student changes their residential address and only the Library office updates their local file while the Fee office does not, the university now holds two contradictory addresses for the same student. This occurs due to uncoordinated duplicate storage without centralized integrity control.

Q60 Marks: 3M ★ Repeated 2x CCSU: 2020, 2017

What is a Data Dictionary (System Catalog)?

Definition: A **Data Dictionary** is a reserved set of read-only system tables maintained by the DBMS that stores metadata (data about data), including schemas, table descriptions, column domains, integrity constraints, indexes, user privileges, and storage allocations.

Q61 Marks: 3M CCSU Exam CCSU: 2024

What are the advantages of using DBMS?

Key Advantages:

1. **Minimal Redundancy:** Normalization ensures data is stored in one primary location.
2. **Data Integrity:** Automatic enforcement of domain and referential integrity constraints.
3. **Concurrency Control:** Multi-user simultaneous access without collisions via locking protocols.
4. **Crash Recovery:** WAL (Write-Ahead Logging) and checkpoints guarantee durability after hardware failure.
5. **Data Independence:** Separates user views from low-level storage modifications.

Q62 Marks: 3M CCSU Exam CCSU: 2024

What is RDBMS?

Definition: A **Relational Database Management System (RDBMS)** is a DBMS based on E.F. Codd's Relational Model (1970). Data is organized into 2D tables (relations) consisting of rows (tuples) and columns (attributes), manipulated using SQL and relational algebra. Examples: Oracle, MySQL, PostgreSQL, SQL Server.

Q63 Marks: 3M CCSU Exam CCSU: 2023

List any five applications of DBMS.

Applications:

1. **Banking:** Customer accounts, loans, ATM transactions, fund transfers.
2. **Airlines & Railways:** Real-time reservation systems, flight schedules, ticketing.
3. **Universities & Colleges:** Student admissions, course registrations, grades, fee accounts.
4. **E-Commerce & Retail:** Online product catalogs, shopping carts, billing, inventory control.
5. **Telecommunications:** Call records, monthly billing, subscriber information, network routing.

Q64 Marks: 3M CCSU Exam CCSU: 2023

What are Attributes? Give examples.

Definition: An **Attribute** is a descriptive property or characteristic that describes an entity in an entity set.
Examples for Entity STUDENT:

- *Simple Attribute:* `RollNo`, `Age`
- *Composite Attribute:* `FullName` (composed of `FirstName`, `LastName`)
- *Multivalued Attribute:* `PhoneNumbers`, `Hobbies`
- *Derived Attribute:* `Age` (computed from `DateOfBirth` and current date)

Q65 Marks: 3M CCSU Exam CCSU: 2020

Define DML (Data Manipulation Language).

Definition: Data Manipulation Language (DML) provides constructs for accessing, inserting, updating, and querying stored database records. Classified into *Procedural DML* (user specifies what data and how to retrieve it) and *Non-Procedural / Declarative DML* (SQL, where user specifies only what data is needed).

Q66 Marks: 3M CCSU Exam CCSU: 2020

Give examples of Simple and Composite Attributes of an entity.

Examples:

- **Simple (Atomic) Attribute:** Cannot be divided into subparts (e.g., `RollNo = 101`, `Age = 21`, `Salary = 50000`).
- **Composite Attribute:** Can be divided into smaller sub-components with independent meanings (e.g., `Address` divided into `HouseNo`, `Street`, `City`, `Pincode`; or `Name` into `FirstName`, `LastName`).

Q67 Marks: 3M CCSU Exam CCSU: 2019

Name the three levels of Data Abstraction.

Three Levels:

1. **Physical Level (Lowest):** Describes how data is stored on disk blocks.
2. **Logical / Conceptual Level (Middle):** Describes what data is stored and relationships among data.
3. **View / External Level (Highest):** Describes customized views for specific user groups.

Q68 Marks: 3M CCSU Exam CCSU: 2019

Distinguish between Functional Dependency and Multivalued Dependency.

Comparison:

- **Functional Dependency (X → Y):** For any two tuples with identical X values, their Y values must be identical (single-valued dependency). Basis of 2NF, 3NF, BCNF.
- **Multivalued Dependency (X ↠ Y):** The presence of (x, y_1, z_1) and (x, y_2, z_2) implies the presence of (x, y_1, z_2) and (x, y_2, z_1), representing independent multi-valued facts. Basis of 4NF.

Q69 Marks: 3M CCSU Exam CCSU: 2018

What do you mean by Relational Database? Describe.

Definition: A **Relational Database** is a collection of data items organized into formal tables (relations) of rows and columns based on first-order predicate logic, where relations are linked together via primary and foreign key constraints.

Q70 Marks: 3M CCSU Exam CCSU: 2018

Briefly explain the terms Security and Authorization.

Definitions:

- **Database Security:** Protection of the database against unauthorized access, malicious attacks, corruption, or destruction.

- **Authorization:** The mechanism that defines and enforces what actions (e.g., SELECT, INSERT, UPDATE, DROP) a validated user or role is permitted to perform on specific database objects.

Q71 Marks: 3M CCSU Exam CCSU: 2017

Define Tuple, Relation, Attribute, and Domain.

Definitions:

- **Relation:** A two-dimensional table of records.
- **Tuple:** A single row in a relation representing an individual entity instance.
- **Attribute:** A named column describing a specific property.
- **Domain:** The set of allowable atomic values for an attribute.

Q72 Marks: 3M CCSU Exam CCSU: 2017

What are Cardinality and Participation Constraints in ER Modeling?

Definitions:

- **Cardinality Ratio:** Specifies the maximum number of relationship instances that an entity can participate in (1:1, 1:N, M:N).
- **Participation Constraint:** Specifies whether existence of an entity depends on its being related to another entity: *Total* (double line) or *Partial* (single line).

Q73 Marks: 3M CCSU Exam CCSU: 2016

What is a Weak Entity Set? How is it identified?

Definition: An entity set lacking sufficient attributes to form a primary key. Identified using a *discriminator (partial key)* combined with the primary key of its identifying strong entity set.

Q74 Marks: 3M CCSU Exam CCSU: 2016

Differentiate between Natural Join and Theta Join.

Differences: Theta Join (R \bowtie_θ S) combines rows based on any arbitrary condition. Natural Join (R \bowtie S) automatically equates attributes having identical names and eliminates duplicate columns.

Q75 Marks: 3M CCSU Exam CCSU: 2016

Explain COMMIT and ROLLBACK commands in SQL.

TCL Commands:

- `COMMIT`: Permanently saves all transaction modifications to disk.
- `ROLLBACK`: Reverts all uncommitted changes back to the start of the transaction or savepoint.

Q76 **Marks: 3M** CCSU Exam **CCSU: 2015**

What do you understand by Hashing? Explain with example.

Definition: Direct access technique mapping search keys to physical bucket addresses via hash function h(k) = k ±od m for O(1) lookups.

Q77 **Marks: 3M** CCSU Exam **CCSU: 2015**

What are Nested SQL Queries? Explain with example.

Definition: A query enclosed inside another SQL query, passed via `WHERE` or `HAVING` clauses using operators like `IN`, `EXISTS`, `ALL`, `ANY`.

```
SELECT EmpName FROM Employee
WHERE Salary > (SELECT AVG(Salary) FROM Employee);
```

Q78 **Marks: 3M** CCSU Exam **CCSU: 2014**

Differentiate between Sequential and Random Access in files.

Differences: Sequential access reads records consecutively in order from the beginning of the file (O(N)). Random (direct) access jumps immediately to any record block by its calculated address or key index (O(1)).

Q79 **Marks: 3M** CCSU Exam **CCSU: 2013**

What is Data Redundancy?

Definition: Unnecessary duplication of identical data items across multiple database files, causing wasted storage and risk of inconsistency.

Q80 **Marks: 3M** CCSU Exam **CCSU: 2013**

What is Relational Data Model?

Definition: A data model representing data in terms of mathematical relations (2D tables) composed of rows (tuples) and columns (attributes).

BCA-502 | Subject Chapter 2 of 4 | 50 Unique Master Questions

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SECTION C • LONG ANSWER QUESTIONS (15 MARKS / COMPREHENSIVE SOLUTIONS) 22 Questions

Q1 Marks: 15M / 3M / 7.5M 🔥 Repeated 9x CCSU: 2025, 2024, 2023, 2021, 2020, 2019, 2018, 2017, 2016

Explain Applet Life Cycle. Write a program to demonstrate a simple Java applet to display an image. Compare Applets over HTML.

Applet Image Display Code:

```
import java.applet.*;
import java.awt.*;

/* <applet code="ImageApplet.class" width="400" height="300"></applet> */
public class ImageApplet extends Applet {
    Image img;
    public void init() {
        img = getImage(getDocumentBase(), "college_logo.png");
    }
    public void paint(Graphics g) {
        g.drawImage(img, 20, 20, this);
        g.drawString("CCSU BCA V Semester", 20, 260);
    }
}
```

Comparison over HTML: Applets provide dynamic client-side computation, graphical interactivity, and local state management that static HTML cannot natively provide.

Q2 Marks: 15M / 3M / 7.5M 🔥 Repeated 9x CCSU: 2025, 2024, 2023, 2021, 2020, 2019, 2018, 2017, 2016

Explain Servlet Life Cycle and compare it with JSP Life Cycle. Write a complete HttpServlet code example.

Servlet Life Cycle Methods:

1. `init(ServletConfig config)` : Loads servlet class, creates instance, and initializes parameters (called once).
2. `service(HttpServletRequest req, HttpServletResponse res)` : Dispatches to `doGet()` or `doPost()` for every incoming HTTP request.
3. `destroy()` : Invoked when container unloads the servlet, releasing database and memory resources.

Servlet Implementation Example:

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class WelcomeServlet extends HttpServlet {
    public void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        String user = request.getParameter("username");
        out.println("<html><body>");
        out.println("<h2>Welcome, " + user + "!</h2>");
        out.println("<p>Server Response Time: " + new java.util.Date() + "</p>");
        out.println("</body></html>");
    }
}
```

Q3 **Marks: 15M / 3M / 7.5M** 🔥 **Repeated 7x** **CCSU: 2025, 2023, 2020, 2019, 2018, 2017, 2016**

Explain Exception Handling mechanism in Java with keywords: try, catch, finally, throw, and throws.

Keywords & Mechanism:

- `try`: Encloses code that may potentially throw an exception.
- `catch`: Catches and handles specific exception types.
- `finally`: Guaranteed cleanup block executed after try/catch.
- `throw`: Explicitly throws an exception object (e.g., `throw new ArithmeticException("Divide by zero");`).
- `throws`: Declares exceptions in method signatures for callers to handle.

Q4 **Marks: 15M / 3M / 7.5M** 🔥 **Repeated 6x** **CCSU: 2025, 2020, 2019, 2018, 2017, 2016**

What is the difference between Process-based and Thread-based multitasking in Java?

Comparison:

Feature	Process-Based Multitasking	Thread-Based Multitasking
Granularity	Heavyweight; each process has its own separate address space.	Lightweight; threads share the same address space in a single process.
Context Switching	High overhead; managed by operating system kernel.	Low overhead; switches quickly within the JVM.
Communication	Requires Inter-Process Communication (IPC, sockets, pipes).	Direct shared memory, wait(), notify(), synchronized blocks.

Q5 **Marks: 15M / 3M / 7.5M** 🔥 **Repeated 6x** **CCSU: 2025, 2023, 2020, 2018, 2017, 2016**

Define Thread in Java. Explain how to create threads using (a) Thread class and (b) Runnable interface. Write a complete multithreaded program.

Multithreading in Java:

```
// Approach 1: Extending Thread class
class NumberPrinter extends Thread {
    public void run() {
        for(int i = 1; i <= 5; i++) {
            System.out.println("Thread 1: " + i);
            try { Thread.sleep(500); } catch(Exception e) {}
        }
    }
}

// Approach 2: Implementing Runnable interface
class LetterPrinter implements Runnable {
    public void run() {
        for(char c = 'A'; c <= 'E'; c++) {
            System.out.println("Thread 2: " + c);
            try { Thread.sleep(500); } catch(Exception e) {}
        }
    }
}

public class MultiThreadDemo {
    public static void main(String[] args) {
        NumberPrinter t1 = new NumberPrinter();
        Thread t2 = new Thread(new LetterPrinter());
        t1.start(); // Spawns new thread
        t2.start();
    }
}
```

Q6 **Marks: 15M / 3M / 7.5M** 🔥 **Repeated 4x** **CCSU: 2024, 2023, 2018, 2016**

What are Packages in Java? How is a user-defined package created and imported? Explain with complete code example.

Creating & Importing User-Defined Package:

```
// File: mypack/Calculator.java
package mypack;

public class Calculator {
    public int add(int a, int b) { return a + b; }
    public int multiply(int a, int b) { return a * b; }
}

// File: TestPackage.java
import mypack.Calculator;

public class TestPackage {
    public static void main(String[] args) {
        Calculator calc = new Calculator();
        System.out.println("Sum: " + calc.add(10, 20));
        System.out.println("Product: " + calc.multiply(10, 20));
    }
}
```

Q7 **Marks: 15M** ★ **Repeated 3x** **CCSU: 2025, 2017, 2016**

What is JDBC? Explain the Four Types of JDBC Drivers with architectural diagrams and characteristics.

Definition: Java Database Connectivity (JDBC) is an industry-standard Java API that enables Java programs to interact with relational databases using SQL queries.

Four Types of JDBC Drivers:

1. **Type-1: JDBC-ODBC Bridge Driver:** Translates JDBC calls to ODBC calls. Requires ODBC driver installation on every client machine. Obsolete and slow.
2. **Type-2: Native-API / Partially Java Driver:** Converts JDBC calls directly into native C/C++ vendor database client libraries (e.g., Oracle OCI). Requires client-side installation.
3. **Type-3: Network Protocol / All-Java Middleware Driver:** Translates JDBC calls into an intermediate server middleware protocol, which then talks to multiple databases. Flexible for 3-tier architectures.
4. **Type-4: Thin Driver / Direct-to-Database Pure Java Driver:** Converts JDBC calls directly into vendor-specific database network protocols via TCP/IP sockets. 100% pure Java, zero client installation, highest performance (Modern standard!).

Q8 **Marks: 15M** ★ Repeated 3x **CCSU: 2025, 2024, 2017**

What are Layout Managers in AWT/Swing? Explain BorderLayout, FlowLayout, and GridLayout with examples.

Layout Managers: Automatically position, size, and rearrange GUI components inside containers regardless of screen resolution or OS window scaling.

- **FlowLayout:** Arranges components sequentially from left to right, wrapping to the next line when space runs out. Default for `Panel` and `Applet`.
- **BorderLayout:** Divides container into 5 regions: `NORTH`, `SOUTH`, `EAST`, `WEST`, and `CENTER`. Default for `Frame` and `JFrame`.
- **GridLayout:** Arranges components into a rectangular grid of equal-sized rows and columns.

Q9 **Marks: 15M** ★ Repeated 3x **CCSU: 2024, 2019, 2018**

What is JDBC? Write a complete JDBC program to insert and display records of employees using PreparedStatement.

Complete PreparedStatement JDBC Program:

```
import java.sql.*;

public class EmployeeJDBC {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/company_db";
        String user = "root";
        String password = "password";

        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection conn = DriverManager.getConnection(url, user, password);

            // 1. Insert record using PreparedStatement
            String insertSql = "INSERT INTO Employee(EmpID, Name, Salary) VALUES(?, ?, ?)";
            PreparedStatement psInsert = conn.prepareStatement(insertSql);
            psInsert.setInt(1, 101);
            psInsert.setString(2, "Rahul Sharma");
            psInsert.setDouble(3, 55000.00);
            int rows = psInsert.executeUpdate();
            System.out.println("Inserted rows: " + rows);

            // 2. Display all employee records
            String selectSql = "SELECT EmpID, Name, Salary FROM Employee";
            Statement stmt = conn.createStatement();
            ResultSet rs = stmt.executeQuery(selectSql);

            System.out.println("\n--- EMPLOYEE RECORDS ---");
            while(rs.next()) {
                System.out.println(rs.getInt("EmpID") + "\t" +
                                   rs.getString("Name") + "\t" +
                                   rs.getDouble("Salary"));
            }
            conn.close();
        } catch(Exception e) {
            e.printStackTrace();
        }
    }
}
```

Q10 **Marks: 15M / 22.5M** **★ Repeated 2x** **CCSU: 2021, 2018**

Explain AWT Event Handling with Delegation Event Model. Write a program to handle Button click events.

Delegation Event Model: An event source generates an event and sends it to registered event listeners implementing listener interfaces (e.g., `ActionListener`).

Q11 **Marks: 15M** **★ Repeated 2x** **CCSU: 2024, 2020**

Discuss JSP in detail. What are JSP Directives? Explain various types of directives with examples.

JSP Directives (`<%@ directive attribute="value" %>`):

1. **page Directive:** Controls page-level settings: `<%@ page import="java.util.*, java.sql.*" contentType="text/html" errorPage="error.jsp" %>`
2. **include Directive:** Inserts static files at translation time: `<%@ include file="header.jsp" %>`
3. **taglib Directive:** Imports custom tag libraries (JSTL): `<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c" %>`

Q12 **Marks: 15M** ★ Repeated 2x **CCSU: 2023, 2020**

What is JDBC? Explain JDBC Architecture and execute a database SELECT query.

JDBC Architecture: Application → JDBC API → DriverManager → JDBC Driver → Database. Complete sample SELECT query program.

Q13 **Marks: 15M** ★ Repeated 2x **CCSU: 2021, 2016**

Explain Client-Server Socket Programming in Java with complete code.

Socket Programming: `ServerSocket` on server side and `Socket` on client side transmitting data over TCP streams using `DataInputStream` and `DataOutputStream`.

Q14 **Marks: 22.5M** CCSU Exam **CCSU: 2021**

Explain JDBC Architecture and write a program to update employee salaries based on department.

Program:

```
import java.sql.*;

public class UpdateSalaryDemo {
    public static void main(String[] args) {
        try {
            Connection con = DriverManager.getConnection("jdbc:mysql://localhost:3306/db", "root", "pass");
            PreparedStatement ps = con.prepareStatement("UPDATE Employee SET Salary = Salary * 1.10 WHERE Dept
= ?");
            ps.setString(1, "IT");
            int rows = ps.executeUpdate();
            System.out.println("Salaries updated for " + rows + " employees.");
            con.close();
        } catch(Exception e) { e.printStackTrace(); }
    }
}
```

Q15 **Marks: 15M** CCSU Exam **CCSU: 2025**

Write an HTML code to create a User Registration Web Form containing: Username, Date of Birth, Address, Gender (Radio buttons), Email-ID, Mobile Number, and Submit/Reset buttons.

HTML Code:

```
<!DOCTYPE html>
<html>
<head>
    <title>User Registration Form</title>
</head>
<body>
    <h2>Student / User Registration Form</h2>
    <form action="register" method="POST">
        <table cellpadding="6">
            <tr>
                <td>User Name:</td>
                <td><input type="text" name="username" required /></td>
            </tr>
            <tr>
                <td>Date of Birth:</td>
                <td><input type="date" name="dob" required /></td>
            </tr>
            <tr>
                <td>Address:</td>
                <td><textarea name="address" rows="3" cols="30"></textarea></td>
            </tr>
            <tr>
                <td>Gender:</td>
                <td>
                    <input type="radio" name="gender" value="Male" /> Male
                    <input type="radio" name="gender" value="Female" /> Female
                </td>
            </tr>
            <tr>
                <td>Email-ID:</td>
                <td><input type="email" name="email" required /></td>
            </tr>
            <tr>
                <td>Mobile Number:</td>
                <td><input type="tel" name="mobile" pattern="[0-9]{10}" required /></td>
            </tr>
            <tr>
                <td colspan="2" align="center">
                    <input type="submit" value="Register Now" />
                    <input type="reset" value="Clear Form" />
                </td>
            </tr>
        </table>
    </form>
</body>
</html>
```

Q16 **Marks: 15M** CCSU Exam **CCSU: 2024**

What are Packages? Explain Java API Packages. List any five methods of the String class with examples.

String Class Methods:

1. `length()` : Returns character count → `"Hello".length() == 5`.
2. `charAt(int index)` : Returns character at index → `"Hello".charAt(1) == 'e'`.
3. `substring(int begin, int end)` : Extracts substring → `"Java".substring(0, 2) == "Ja"`.
4. `toUpperCase()` : Converts to uppercase → `"bca".toUpperCase() == "BCA"`.
5. `equals(Object another)` : Compares string contents → `"Java".equals("Java") == true`.

Q17 Marks: 15M CCSU Exam CCSU: 2023

Explain HTML form controls and write code for an Admission Form.

Admission Form: Includes text fields, dropdown selects, checkboxes, text areas, and submit buttons.

Q18 Marks: 15M CCSU Exam CCSU: 2023

Explain JSP implicit objects and session tracking in Web Applications.

Implicit Objects: `request`, `response`, `session`, `application`, `out`, `config`, `pageContext`, `page`, `exception`. Session tracking via Cookies, URL Rewriting, and HttpSession.

Q19 Marks: 15M CCSU Exam CCSU: 2020

Write HTML code for a complete Web Form with Table formatting and Validation.

HTML Implementation: Form markup with input types, select dropdowns, radio buttons, and table alignments.

Q20 Marks: 15M CCSU Exam CCSU: 2020

Explain Servlets, HttpServletRequest, and HttpServletResponse with complete GET/POST demonstration.

Servlet Processing: Complete explanation of HTTP request parsing, header inspection, and HTML output streaming.

Q21 Marks: 15M CCSU Exam CCSU: 2019

Explain Event Handling in Java AWT/Swing with MouseListener and ActionListener.

Event Handling: Source → Event Object → Listener interface implementation.

Q22 Marks: 15M CCSU Exam CCSU: 2019

Write HTML and CSS code to design a responsive institutional web page.

Web Design: Clean HTML markup with external/internal CSS styling, tables, lists, and navigation bar.

SECTION B • MEDIUM ANSWER QUESTIONS (7.5 MARKS / CORE CONCEPTUAL SOLUTIONS)

7 Questions

Q23 Marks: 7.5M ★ Repeated 3x CCSU: 2024, 2023, 2019

Explain the concepts of Inheritance in Java and its various types.

Inheritance: The mechanism where a subclass derives attributes and behaviors from a superclass using `extends`.
Types in Java:

1. **Single Inheritance:** Class B extends Class A.
2. **Multilevel Inheritance:** Class C extends Class B, which extends Class A.
3. **Hierarchical Inheritance:** Class B and Class C both extend Class A.

4. **Multiple Inheritance:** Achieved through **Interfaces** (Java does not support multiple class inheritance to avoid the Diamond Problem).

Q24 **Marks: 3M / 7.5M** CCSU Exam **CCSU: 2024**

Explain TCP/IP Client Socket. Write constructors and methods used to create client sockets.

Client Socket (java.net.Socket):

- **Constructors:**
  - `Socket(String host, int port)` — Establishes TCP stream connection.
  - `Socket(InetAddress address, int port)`
- **Methods:**
  - `getInputStream()` — Returns InputStream for reading server responses.
  - `getOutputStream()` — Returns OutputStream for sending data.
  - `close()` — Gracefully closes the TCP connection.

Q25 **Marks: 7.5M** CCSU Exam **CCSU: 2021**

What do you mean by a Package in Java? State the different Access Specifiers available in Java.

Access Specifiers:

Access Specifier	Same Class	Same Package	Subclass (Diff Pkg)	World (Everywhere)
`private`	Yes	No	No	No
`default` (package-private)	Yes	Yes	No	No
`protected`	Yes	Yes	Yes (via inheritance)	No
`public`	Yes	Yes	Yes	Yes

Q26 **Marks: 7.5M** CCSU Exam **CCSU: 2021**

What is Exception? Give different types of exceptions that occur during runtime. Why handle exceptions?

Explanation: An **Exception** is an abnormal runtime event that disrupts the normal flow of program instructions. Types: Checked (`IOException` , `SQLException`) and Unchecked/Runtime (`ArithmeticException` , `NullPointerException` , `ArrayIndexOutOfBoundsException`). Handled to maintain graceful execution and avoid abrupt program termination.

Q27 **Marks: 7.5M** CCSU Exam **CCSU: 2021**

Explain Thread Priority. What are default values? Write a program to define two threads where one prints 1 to 10 and the other prints 10 to 1.

Thread Priorities: Integer scale from 1 (`MIN_PRIORITY`) to 10 (`MAX_PRIORITY`), with default 5 (`NORM_PRIORITY`).
Program:

```
class ForwardPrinter extends Thread {
    public void run() {
        for(int i = 1; i <= 10; i++) {
            System.out.println("Forward: " + i);
            try { Thread.sleep(200); } catch(Exception e) {}
        }
    }
}

class BackwardPrinter extends Thread {
    public void run() {
        for(int j = 10; j >= 1; j--) {
            System.out.println("Backward: " + j);
            try { Thread.sleep(200); } catch(Exception e) {}
        }
    }
}

public class PriorityThreadDemo {
    public static void main(String[] args) {
        ForwardPrinter t1 = new ForwardPrinter();
        BackwardPrinter t2 = new BackwardPrinter();
        t1.setPriority(Thread.MAX_PRIORITY);
        t2.setPriority(Thread.MIN_PRIORITY);
        t1.start();
        t2.start();
    }
}
```

Q28 Marks: 7.5M CCSU Exam CCSU: 2017

Explain Packages and Access Protection in Java.

Access Protection: Detailed comparison of private, default, protected, and public scopes.

Q29 Marks: 7.5M CCSU Exam CCSU: 2016

What do you mean by Constructor? Explain Constructor Overloading.

Constructor Overloading: Defining multiple constructors in a single class with different parameter lists.

SECTION A • SHORT ANSWER QUESTIONS (3 MARKS / PRECISE DEFINITIVE SOLUTIONS)

21 Questions

Q30 Marks: 3M 🔥 Repeated 4x CCSU: 2025, 2019, 2018, 2016

Explain Garbage Collection in Java.

Definition: Java features automatic memory management via the Garbage Collector (GC), a background daemon thread that periodically scans the Heap memory and deallocates memory occupied by unreachable (unreferenced) objects, preventing memory leaks without manual pointer freeing.

Q31 Marks: 3M ★ Repeated 3x CCSU: 2025, 2019, 2017

What is Bytecode in Java? Why does it enable platform independence?

Explanation: Java source code (`.java`) is compiled by `javac` into an intermediate machine-independent format called **Bytecode** (stored in `.class` files). Any platform equipped with a standard Java Virtual Machine (JVM) interprets and executes this bytecode natively, achieving "Write Once, Run Anywhere" (WORA).

Q32 Marks: 3M ★ Repeated 2x CCSU: 2025, 2018

Explain the 'super' keyword in Java with usages.

Usage of 'super': A reference variable used in a subclass to refer directly to its immediate parent class object:

1. **Invoke Parent Constructor:** `super(args);` — must be the first line in subclass constructor.
2. **Access Parent Variables:** `super.variableName` — accesses shadowed parent instance fields.
3. **Call Parent Methods:** `super.methodName()` — invokes overridden parent class methods.

Q33 Marks: 3M ★ Repeated 2x CCSU: 2024, 2020

Discuss Primitive Data Types in Java.

Java Data Types:

- **Integer Types:** `byte` (1 byte), `short` (2 bytes), `int` (4 bytes), `long` (8 bytes).
- **Floating-Point:** `float` (4 bytes, IEEE 754), `double` (8 bytes).
- **Character:** `char` (2 bytes, 16-bit Unicode character).
- **Boolean:** `boolean` (represents true or false).

Q34 Marks: 3M ★ Repeated 2x CCSU: 2023, 2017

What is Method Overloading vs Method Overriding?

Comparison: Overloading occurs in the same class with identical method names but different parameter lists (compile-time polymorphism). Overriding occurs in subclasses with identical method signatures and return types (runtime dynamic dispatch).

Q35 Marks: 3M ★ Repeated 2x CCSU: 2020, 2017

What is the purpose of the 'static' keyword in Java?

Explanation: Marks class-level members (variables, methods, blocks) shared by all instances, callable without creating an object of the class (e.g., `ClassName.staticMethod()`).

Q36 Marks: 3M ★ Repeated 2x CCSU: 2019, 2018

What is a Final Variable? Give an example.

Definition: A variable declared with the `final` keyword becomes a constant whose value cannot be reassigned once initialized:

```
final double PI = 3.14159;
// PI = 3.14; // Compile-time error: cannot assign a value to final variable
```

Q37 **Marks: 3M** CCSU Exam **CCSU: 2024**

Explain the Anchor and Table tags in HTML.

HTML Tags:

- `<a href="url">Link Text</a>`: Creates a hyperlink to navigate between web pages or download files.
- `<table><tr><th>...</th><td>...</td></tr></table>`: Renders structured tabular data using rows (`<tr>`), headers (`<th>`), and data cells (`<td>`).

Q38 **Marks: 3M** CCSU Exam **CCSU: 2024**

Briefly explain Internet Concepts: URL, HTTP, and Web Server.

Concepts:

- **URL (Uniform Resource Locator):** Global address of a resource on the web (e.g., `https://www.example.com/index.html`).
- **HTTP:** HyperText Transfer Protocol, a stateless client-server protocol running over TCP port 80/443 for web communication.
- **Web Server:** Hardware/software that serves web resources to clients (Apache, Nginx, Tomcat).

Q39 **Marks: 3M** CCSU Exam **CCSU: 2023**

How is an Object created in Java?

Object Creation: Uses the `new` operator:

```
ClassName obj = new ClassName();
```

Steps: (1) *Declaration* (associates variable name with object type), (2) *Instantiation* (`new` allocates memory on the Heap), (3) *Initialization* (invokes the constructor).

Q40 **Marks: 3M** CCSU Exam **CCSU: 2023**

Compare JDK, JRE, and JVM.

Relationship:

- **JVM (Java Virtual Machine):** The runtime engine that interprets and executes bytecode.
- **JRE (Java Runtime Environment):** Contains JVM + Java Standard Class Libraries; sufficient to run Java programs.
- **JDK (Java Development Kit):** Contains JRE + Development Tools (`javac` , `javadoc` , `jar` , `debugger`); needed to write and compile programs.

Q41 Marks: 3M CCSU Exam CCSU: 2023

Difference between JDBC and ODBC.

Differences: ODBC is a C/C++ platform-dependent API designed originally for Windows. JDBC is a platform-independent pure Java API tailored for Java applications and cross-platform RDBMS connectivity.

Q42 Marks: 3M CCSU Exam CCSU: 2020

Explain why Java is considered the best language for Internet Applications.

Reasons: Platform independence via bytecode/JVM, built-in security (sandbox security manager), robust memory management, native multithreading, and extensive networking APIs (`java.net`).

Q43 Marks: 3M CCSU Exam CCSU: 2020

What are Wrapper Classes in Java?

Explanation: Classes that wrap primitive data types into object representations (`Integer` , `Double` , `Boolean`) enabling autoboxing and unboxing for collections.

Q44 Marks: 3M CCSU Exam CCSU: 2019

List the four fundamental principles of Object-Oriented Programming.

OOP Principles: Encapsulation, Abstraction, Inheritance, and Polymorphism.

Q45 Marks: 3M CCSU Exam CCSU: 2019

Define Constructor in Java.

Definition: A special method called automatically when an object is instantiated, having the same name as the class and no return type.

Q46 Marks: 3M CCSU Exam CCSU: 2018

What is an Abstract Method?

Definition: A method declared without a body (implementation) using the `abstract` keyword, terminating with a semicolon. Must be implemented by subclasses.

Q47 Marks: 3M CCSU Exam CCSU: 2018

What is a Layout Manager in Java AWT?

Definition: An interface/class that manages the automatic positioning and sizing of GUI components within a container window.

Q48 **Marks: 3M** CCSU Exam **CCSU: 2017**

How will you create hyperlinks in a web document? Write the code also.

HTML Hyperlink:

```
<a href="https://www.ccsuniversity.ac.in" target="_blank">Visit CCSU Official Portal</a>
```

Q49 **Marks: 3M** CCSU Exam **CCSU: 2017**

What is a Constructor? List its characteristics.

Characteristics: Same name as class, no return type (not even void), automatically called on `new`, can be overloaded.

Q50 **Marks: 3M** CCSU Exam **CCSU: 2016**

What is Byte-code in JAVA? Explain.

Definition: The intermediate machine instruction set generated by `javac`, executed by the JVM to provide platform independence.

BCA-503 | Subject Chapter 3 of 4 | 48 Unique Master Questions

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SECTION C • LONG ANSWER QUESTIONS (15 MARKS / COMPREHENSIVE SOLUTIONS) 13 Questions

Q1 Marks: 15M / 3M / 7.5M Repeated 9x CCSU: 2025, 2024, 2023, 2021, 2020, 2019, 2018, 2017, 2016

Explain the ISO/OSI Reference Model in detail. Describe the primary functions of each of the seven layers.

Seven Layers of OSI Model:

1. **Physical Layer (Layer 1):** Transmission of raw unstructured bit streams over physical media (voltages, connectors, cables).
2. **Data Link Layer (Layer 2):** Framing, physical addressing (MAC), flow control, error control (CRC), and media access control.
3. **Network Layer (Layer 3):** Logical addressing (IP), routing (Dijkstra, distance vector), packet forwarding, and fragmentation.
4. **Transport Layer (Layer 4):** End-to-end process-to-process delivery (ports), segmentation and reassembly, connection control (TCP handshake), reliable flow control.
5. **Session Layer (Layer 5):** Dialogue control (simplex, half/full duplex), session checkpointing, synchronization tokens.
6. **Presentation Layer (Layer 6):** Data translation, character code conversion (ASCII/EBCDIC), compression, encryption/decryption (SSL/TLS).
7. **Application Layer (Layer 7):** Network virtual terminals, file transfer (FTP), web browsing (HTTP), email services (SMTP).

Q2 Marks: 15M / 7.5M Repeated 8x CCSU: 2025, 2024, 2023, 2021, 2020, 2019, 2017, 2016

Explain Congestion Control Mechanisms in Computer Networks: Leaky Bucket and Token Bucket Algorithms with comparative analysis.

Traffic Shaping Algorithms:

- **Leaky Bucket:** Discharges data at a strictly constant, uniform output rate regardless of incoming burstiness. If incoming burst exceeds bucket buffer capacity, excess packets overflow and drop. Smooths traffic completely.
- **Token Bucket:** Generates tokens at a constant rate up to bucket capacity. Allows transmission of bursty traffic as long as sufficient tokens exist, but limits sustained rate. More flexible and widely used in modern QoS.

Q3 Marks: 15M / 22.5M / 7.5M Repeated 7x CCSU: 2025, 2021, 2020, 2019, 2018, 2017, 2016

Explain Sliding Window Protocols: Stop-and-Wait ARQ, Go-Back-N ARQ, and Selective Repeat ARQ with efficiency analysis.

Flow and Error Control Protocols:

- **Stop-and-Wait ARQ:** Sender transmits 1 frame, starts timer, and waits for ACK before sending next. Efficiency: \eta = (1 / 1 + 2a) where a = T_p / T_t. Poor for high bandwidth-delay links.
- **Go-Back-N ARQ:** Sender window size W_s = 2^m - 1, receiver window W_r = 1. Sender can transmit multiple frames before receiving ACK. On loss, retransmits lost frame and all subsequent frames. Cumulative ACKs.
- **Selective Repeat ARQ:** Sender window W_s = 2^{m-1}, receiver window W_r = 2^{m-1}. Receiver accepts out-of-order frames within window. Sender retransmits *only the specific corrupted/timed-out frames* (highest throughput!).

Q4 **Marks: 15M / 3M / 7.5M** **Repeated 7x** **CCSU: 2025, 2024, 2020, 2019, 2018, 2017, 2016**

Compare Circuit Switching, Packet Switching, and Message Switching.

Comparison:

Feature	Circuit Switching	Packet Switching	Message Switching
Dedicated Path	Yes (setup before transfer).	No (packets routed dynamically).	No (store-and-forward per hop).
Bandwidth	Reserved/wasted if idle.	Dynamically shared on-demand.	Dynamically shared.
Delay	Setup delay high; transmission delay low.	Jitter/variable queuing delay.	High storage delay at each switch.
Modern Use	PSTN landline telephone.	Internet (IP networks).	Early telegraph, postal email.

Q5 **Marks: 15M / 22.5M / 7.5M** **Repeated 6x** **CCSU: 2025, 2021, 2020, 2019, 2018, 2016**

Explain Routing Algorithms: (a) Distance Vector Routing (Bellman-Ford) and (b) Link State Routing (Dijkstra). Discuss Count-to-Infinity problem.

Routing Algorithms:

1. **Distance Vector Routing:** Each router maintains a routing table containing distance to each destination and next hop. Routers periodically share complete tables with immediate neighbors. Uses Bellman-Ford algorithm. Vulnerable to *Count-to-Infinity problem* when links fail (mitigated by Split Horizon and Poison Reverse).
2. **Link State Routing:** Each router floods Link State Packets (LSP) containing status of its local links to all routers in the entire autonomous system. Every router constructs a global topology map and computes shortest paths using Dijkstra's algorithm (used in OSPF). Fast convergence and no routing loops.

Q6 **Marks: 15M / 7.5M** **Repeated 6x** **CCSU: 2025, 2020, 2019, 2018, 2017, 2016**

Explain Error Detection Techniques: Parity Check, Checksum, and CRC (Cyclic Redundancy Check).

Error Detection:

- **Parity Check:** Appends single parity bit (even or odd parity). Detects only odd numbers of bit errors.
- **Checksum:** Divides data into 16-bit words, adds them using 1's complement arithmetic, and appends inverted sum (used in TCP/IP).
- **Cyclic Redundancy Check (CRC):** Treats bit stream as polynomial coefficients and performs modulo-2 binary division by generator polynomial G(x). Appends remainder (FCS). Detects all single, double, odd, and burst errors up to generator length!

Q7 **Marks: 15M** **Repeated 6x** **CCSU: 2023, 2020, 2019, 2018, 2017, 2016**

Explain Guided Media: Twisted Pair Cable, Coaxial Cable, and Fiber Optic Cable.

Transmission Media: Construction, signal propagation, attenuation characteristics, and network cabling standards (Cat5e, Cat6, Single-mode/Multi-mode fiber).

Q8 **Marks: 15M / 3M / 7.5M** ★ Repeated 3x **CCSU: 2025, 2023, 2016**

What is Subnetting? Give an example.

Definition: Subnetting is the process of partitioning a single physical network into two or more smaller logical subnets by borrowing host bits to create subnet bits, reducing broadcast traffic and optimizing address allocation.
Example: Given Class C network `192.168.1.0/24` with default mask `255.255.255.0`. Borrowing 2 host bits gives subnet mask `255.255.255.192` (/26), creating 2^2 = 4 subnets with 62 usable hosts each:

- Subnet 1: `192.168.1.0` to `192.168.1.63`
- Subnet 2: `192.168.1.64` to `192.168.1.127`
- Subnet 3: `192.168.1.128` to `192.168.1.191`
- Subnet 4: `192.168.1.192` to `192.168.1.255`

Q9 **Marks: 15M** ★ Repeated 2x **CCSU: 2025, 2023**

Write detailed notes on: (a) DNS (Domain Name System), (b) Firewalls and Network Security, (c) SMTP and Email Architecture.

Detailed Notes:

- **DNS:** Hierarchical distributed database mapping hostnames (`www.google.com`) to IP addresses (`142.250.190.46`) over UDP/TCP port 53.
- **Firewalls:** Security barrier controlling inbound/outbound network traffic based on rules. Types: Packet Filtering, Stateful Inspection, Proxy/Application-level gateways.
- **Email Architecture:** User Agent (UA) composes email, Mail Transfer Agent (MTA) transmits using SMTP (port 25/587), and Mail Access Agent (MAA) retrieves email to client via POP3 (port 110) or IMAP (port 143).

Q10 **Marks: 15M** ★ Repeated 2x **CCSU: 2024, 2023**

Explain Error Control in Data Link Layer using CRC with a complete numerical example.

CRC Numerical Example:
Data: `1010000`, Generator: `1001`.
Append 3 zeros: `1010000000`. Perform modulo-2 binary division by `1001` to calculate remainder (FCS). Complete step-by-step arithmetic shown.

Q11 **Marks: 15M / 7.5M** ★ Repeated 2x **CCSU: 2023, 2017**

Explain Dijkstra's Shortest Path Routing Algorithm with a network graph example.

Dijkstra's Algorithm: Greedy shortest path calculation from source router to all other nodes maintaining visited sets and updating tentative distances.

Q12 **Marks: 15M** CCSU Exam **CCSU: 2024**

(a) Explain Shannon Capacity with examples. (b) What is CSMA? List protocols used with CSMA.

(a) Shannon Capacity Theorem: Gives theoretical maximum data rate C of a channel with bandwidth B and Signal-to-Noise Ratio SNR:
C = B \log_2(1 + SNR) bps

Example: For telephone line with B = 3000 Hz and SNR = 3162 (35 dB): C = 3000 \log_2(1 + 3162) ≈ 3000 × 11.62 = 34,860 bps ≈ 34.8 kbps.

(b) CSMA Protocols: 1-Persistent CSMA, Non-Persistent CSMA, p-Persistent CSMA, and CSMA/CD / CSMA/CA.

Q13 **Marks: 15M** CCSU Exam CCSU: 2024

Describe any three: (a) SSL, (b) Modems, (c) Line Configuration, (d) Distortion vs Noise.

Short Notes:

- **SSL (Secure Sockets Layer):** Cryptographic protocol providing end-to-end encryption, authentication, and integrity over TCP.
- **Modems (Modulator-Demodulator):** Converts digital computer signals to analog audio frequencies for transmission over analog lines and vice versa.
- **Line Configuration:** Point-to-Point (dedicated link between two devices) vs Multipoint (multidrop link where multiple devices share a single link).
- **Distortion vs Noise:** Distortion is alteration of signal shape caused by different frequency phase delays in transmission media. Noise is unwanted external thermal/electromagnetic energy added into the signal.

SECTION B • MEDIUM ANSWER QUESTIONS (7.5 MARKS / CORE CONCEPTUAL SOLUTIONS)

10 Questions

Q14 **Marks: 3M / 7.5M** ★ Repeated 2x CCSU: 2025, 2021

Write short notes on: (a) Throughput & Wavelength, (b) Synchronous Protocols, (c) Routers.

Notes:

- **(a) Throughput & Wavelength:** Throughput is the actual measurement of data transferred across a network per second. Wavelength (λ = v / f) is the physical distance between consecutive signal wave peaks in optical communication.
- **(b) Synchronous Protocols:** Continuous transmission of data blocks synchronized to a shared clock, using sync bytes/flags (e.g. HDLC).
- **(c) Routers:** Network Layer (Layer 3) internetworking devices that inspect destination IP addresses and make packet forwarding decisions based on routing tables.

Q15 **Marks: 3M / 7.5M** ★ Repeated 2x CCSU: 2025, 2017

Describe Multiplexing. Explain FDM, TDM, and WDM with diagrams.

Definition: Combining multiple separate data streams into a single composite signal transmitted over a shared transmission medium.
Techniques:

- **FDM (Frequency Division Multiplexing):** Divides channel bandwidth into separate non-overlapping frequency bands with guard bands (Analog signals, radio, cable TV).
- **TDM (Time Division Multiplexing):** Interleaves time slots from multiple sources onto a high-speed digital carrier (Synchronous vs Statistical TDM).
- **WDM (Wavelength Division Multiplexing):** Multiplexes optical signals of differing wavelengths (colors) along a single optical fiber.

Q16 Marks: 3M / 7.5M ★ Repeated 2x CCSU: 2024, 2016

Explain the switching methods in Computer Networks.

Switching Methods: Circuit Switching (dedicated physical communication path established before data transfer), Packet Switching (data split into discrete packets routed dynamically), and Message Switching (entire message stored and forwarded hop-by-hop).

Q17 Marks: 3M / 7.5M ★ Repeated 2x CCSU: 2024, 2023

Explain in brief: (a) Guided Media, (b) Transmission Modes.

(a) Guided Media: Physical physical conductors that guide signals along a specific path: Twisted Pair (UTP/STP), Coaxial Cable, and Optical Fiber.
(b) Transmission Modes:

- *Simplex:* Unidirectional transmission (e.g., keyboard to CPU, TV broadcast).
- *Half-Duplex:* Bidirectional, but only one direction at a time (e.g., Walkie-Talkie).
- *Full-Duplex:* Simultaneous bidirectional transmission (e.g., telephone call).

Q18 Marks: 7.5M CCSU Exam CCSU: 2023

Describe the functions & responsibilities of the Session Layer.

Session Layer (Layer 5):

- **Session Establishment, Maintenance, and Termination:** Manages dialogs between communicating processes.
- **Dialog Control:** Determines whether communication is simplex, half-duplex, or full-duplex.
- **Synchronization / Checkpointing:** Inserts sync tokens into data streams to allow recovery from checkpoints in case of interruptions.

Q19 Marks: 7.5M CCSU Exam CCSU: 2021

Discuss ISDN, its services, and layer structure.

ISDN (Integrated Services Digital Network): Transmits voice, data, and video simultaneously over standard telephone copper lines. Channels: B-channel (64 kbps, user data), D-channel (16 or 64 kbps, control signaling). BRI provides 2B+D (144 kbps); PRI provides 23B+D (1.544 Mbps) or 30B+D (2.048 Mbps).

Q20 Marks: 7.5M CCSU Exam CCSU: 2021

Explain TCP Packet Format (Header Structure) in detail.

TCP Header Fields:

- Source Port (16 bits) and Destination Port (16 bits)
- Sequence Number (32 bits) and Acknowledgment Number (32 bits)
- Header Length / Data Offset (4 bits) and Reserved bits (6 bits)
- Control Flags (6 bits: URG, ACK, PSH, RST, SYN, FIN)
- Window Size (16 bits) for flow control
- Checksum (16 bits) and Urgent Pointer (16 bits)

Q21 Marks: 7.5M CCSU Exam CCSU: 2020

How does DTE-DCE interface work? Discuss the essence of Modems.

DTE-DCE Interface: DTE (Data Terminal Equipment, e.g. computer/router) generates digital data. DCE (Data Circuit-terminating Equipment, e.g. modem) converts signals for transmission across the physical telephone link. Standards include RS-232 / EIA-232.

Q22 Marks: 7.5M CCSU Exam CCSU: 2018

Explain any one protocol used for flow control in noisy channel: Stop-and-Wait ARQ.

Stop-and-Wait ARQ: Sender sends frame with sequence number (0 or 1) and starts timer. If positive ACK received, next frame is sent. If ACK lost or timer expires, frame is retransmitted. Handles lost frames and duplicate frames using 1-bit sequence numbering.

Q23 Marks: 7.5M CCSU Exam CCSU: 2018

Explain Fiber Distributed Data Interface (FDDI) operation.

FDDI: High-speed 100 Mbps token-passing optical fiber network based on dual counter-rotating token rings (Primary and Secondary). If the primary ring breaks, the dual ring self-heals by wrapping around, ensuring fault-tolerant operation.

SECTION A • SHORT ANSWER QUESTIONS (3 MARKS / PRECISE DEFINITIVE SOLUTIONS)

25 Questions

Q24 Marks: 3M 🔥 Repeated 4x CCSU: 2023, 2019, 2018, 2016

What is Network Topology? List advantages and disadvantages of Mesh Topology.

Mesh Topology:

- *Advantages:* Dedicated point-to-point links eliminate traffic congestion; robust fault tolerance (link failure affects only two nodes); excellent privacy.
- *Disadvantages:* Requires massive cabling (N(N-1)/2 links) and expensive multi-port hardware; complex installation.

Q25 Marks: 3M ★ Repeated 3x CCSU: 2019, 2018, 2016

Which device is used for Modulation and Demodulation? Explain its function.

Answer: A **MODEM (Modulator-Demodulator)**. Converts digital computer pulses to analog signals for transmission over analog lines (modulation) and translates received analog waves back into digital bits (demodulation).

Q26 Marks: 3M ★ Repeated 2x CCSU: 2018, 2016

Explain Transmission Modes: Simplex, Half-Duplex, and Full-Duplex.

Modes: Simplex is strictly one-way (keyboard to monitor). Half-Duplex is bidirectional but one direction at a time (walkie-talkie). Full-Duplex is simultaneous two-way transmission (telephone conversation).

Q27 **Marks: 3M** CCSU Exam **CCSU: 2025**

What is a MAC address? How is it different from an IP Address?

Comparison:

Feature	MAC Address (Physical Address)	IP Address (Logical Address)
Layer	Data Link Layer (Layer 2).	Network Layer (Layer 3).
Size & Format	48 bits (6 bytes), hex (e.g., `00:1A:2B:3C:4D:5E`).	32 bits (IPv4, e.g., `192.168.1.1`) or 128 bits (IPv6).
Nature	Hardcoded by manufacturer into NIC; globally fixed.	Dynamically/statically assigned based on network location.

Q28 **Marks: 3M** CCSU Exam **CCSU: 2025**

Differentiate between a Hub and a Switch.

Differences:

- **Hub (Physical Layer - Layer 1):** Broadcast device that repeats every incoming signal across all ports, sharing a single collision domain. Half-duplex, no intelligence.
- **Switch (Data Link Layer - Layer 2):** Intelligent device maintaining a MAC Address Table to forward frames directly to designated destination ports. Dedicated collision domains per port, full-duplex.

Q29 **Marks: 3M** CCSU Exam **CCSU: 2025**

What is CSMA/CD? How does it operate in Ethernet?

Carrier Sense Multiple Access with Collision Detection: A media access protocol used in shared half-duplex Ethernet:

1. **Carrier Sense:** Station listens before transmitting ("listen before talk"). If channel is idle, it begins transmitting.
2. **Collision Detection:** While transmitting, station monitors signal level. If collision occurs, it transmits a *jam signal* to notify all stations.
3. **Backoff:** Stations wait a random backoff period using the Binary Exponential Backoff algorithm before retransmitting.

Q30 **Marks: 3M** CCSU Exam **CCSU: 2024**

What is the need for a Computer Network?

Need for Computer Networks: Resource sharing (printers, storage servers), centralized data management, cost reduction, electronic communication (email, video conferences), and distributed processing power.

Q31 **Marks: 3M** CCSU Exam **CCSU: 2024**

Define HDLC (High-Level Data Link Control).

Definition: HDLC is a bit-oriented synchronous data link layer protocol developed by ISO that provides reliable point-to-point and multipoint communication using frame delimitation (flag byte `01111110` with bit stuffing), error detection, and sliding window flow control.

Q32 Marks: 3M CCSU Exam CCSU: 2024

Explain the concept of Piggybacking.

Definition: Piggybacking is a flow control optimization technique where an acknowledgment (ACK) for received data is delayed and attached into the header of the next outgoing data frame sent back to the sender, eliminating the overhead of transmitting independent acknowledgment packets.

Q33 Marks: 3M CCSU Exam CCSU: 2024

Describe how SMTP protocol is used in E-Mail applications.

SMTP (Simple Mail Transfer Protocol): A push protocol running over TCP port 25 that transfers email messages from a sender's user agent/client to the sender's mail server, and from the sender's mail server to the recipient's mail server across the Internet.

Q34 Marks: 3M CCSU Exam CCSU: 2023

Explain the use of Bridges in Computer Networks.

Definition: A **Bridge** is a Data Link Layer (Layer 2) device that connects two separate LAN segments running the same protocol, inspecting MAC addresses to filter frames and segment collision domains.

Q35 Marks: 3M CCSU Exam CCSU: 2023

Write a short note on E-Mail and its core protocols.

Explanation: Electronic mail architecture consists of User Agents (UAs) and Message Transfer Agents (MTAs). Uses SMTP for sending emails, and POP3 / IMAP for downloading emails to client devices.

Q36 Marks: 3M CCSU Exam CCSU: 2023

Define Bandwidth and Throughput.

Definitions:

- **Bandwidth:** The maximum theoretical data-carrying capacity of a communication channel (bits per second).
- **Throughput:** The actual rate of successfully transmitted data delivered across the network per unit time.

Q37 Marks: 3M CCSU Exam CCSU: 2020

Four channels, each with 100-KHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 KHz between channels?

Solution: N = 4 channels of 100 KHz each.
Number of guard bands required between 4 channels = 4 - 1 = 3.
Each guard band = 10 KHz.
Total minimum bandwidth:
B = (4 × 100) + (3 × 10) = 400 + 30 = **430 KHz**

Q38 Marks: 3M CCSU Exam CCSU: 2020

Discuss Point-to-Point Protocol (PPP).

PPP: A data link layer protocol used to establish a direct connection between two nodes over dial-up, serial cable, or broadband (PPPoE). Provides authentication via PAP/CHAP, framing, and dynamic IP configuration (IPCP).

Q39 Marks: 3M CCSU Exam CCSU: 2020

Differentiate between LAN, MAN, and WAN.

Comparison: LAN covers a room or building (high speed, private ownership). MAN covers a city (cable TV network). WAN covers a country or globe (telecom lines, satellite, Internet).

Q40 Marks: 3M CCSU Exam CCSU: 2020

What is a Collision Domain vs Broadcast Domain?

Definitions: A Collision Domain is a network segment where simultaneous transmissions collide (divided by switches). A Broadcast Domain is a segment receiving all broadcast frames (divided by routers).

Q41 Marks: 3M CCSU Exam CCSU: 2019

Differentiate between Bridges and Repeaters.

Differences: Repeaters operate at the Physical Layer (Layer 1) to regenerate electrical signals without looking at frames. Bridges operate at the Data Link Layer (Layer 2) to inspect MAC addresses and filter traffic between segments.

Q42 Marks: 3M CCSU Exam CCSU: 2019

What are the services provided by the Transport Layer?

Services: Connection-oriented delivery (TCP) vs connectionless datagram delivery (UDP), port-based process addressing, segmentation/reassembly, end-to-end flow control, and error recovery.

Q43 Marks: 3M CCSU Exam CCSU: 2019

What is Shannon Capacity Theorem?

Theorem: C = B \log_2(1 + SNR) gives the theoretical upper bound on data transmission speed in bits per second over a noisy channel.

Q44 Marks: 3M CCSU Exam CCSU: 2018

What is Bit Rate vs Baud Rate?

Definitions: Bit rate is the number of bits transmitted per second (bps). Baud rate is the number of signal unit changes (symbols) per second. Bit Rate = Baud Rate × Bits per Baud.

Q45 **Marks: 3M** CCSU Exam **CCSU: 2017**

A local telephone network is an example of which switching technology?

Answer: Circuit Switching. A dedicated physical communication circuit is established between the calling and called parties through telephone exchanges for the entire duration of the call.

Q46 **Marks: 3M** CCSU Exam **CCSU: 2017**

What is Guided Media? List examples.

Definition: Physical solid transmission media that provide a physical conduit for electromagnetic signals: Twisted Pair Cable, Coaxial Cable, and Fiber Optic Cable.

Q47 **Marks: 3M** CCSU Exam **CCSU: 2017**

Compare TCP and UDP at the Transport Layer.

Comparison: TCP is connection-oriented, reliable (acknowledgments, retransmissions), with flow and congestion control (HTTP, FTP). UDP is connectionless, unreliable, lightweight, and low-latency (DNS, VoIP, video streaming).

Q48 **Marks: 3M** CCSU Exam **CCSU: 2016**

What is Bandwidth of a channel?

Definition: The range of frequencies contained in a signal or passed by a transmission channel: B = f_{max} - f_{min} (measured in Hertz, or bps in digital communication).

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SECTION C • LONG ANSWER QUESTIONS (15 MARKS / COMPREHENSIVE SOLUTIONS) 16 Questions

Q1 Marks: 15M / 22.5M / 3M Repeated 7x CCSU: 2024, 2023, 2022, 2021, 2020, 2019, 2018

Using Runge-Kutta 4th order method, find y(0.2) if dy/dx = x + y^2, given y(0) = 1 and step size h = 0.1.

Solution: We need 2 steps of h = 0.1 to reach x = 0.2.
f(x, y) = x + y^2.
STEP 1: From x_0 = 0, y_0 = 1 to x_1 = 0.1:
k_1 = h f(0, 1) = 0.1(0 + 1) = 0.1000
k_2 = h f(0 + 0.05, 1 + 0.05) = 0.1(0.05 + 1.05^2) = 0.1(0.05 + 1.1025) = 0.11525
k_3 = h f(0 + 0.05, 1 + 0.057625) = 0.1(0.05 + 1.057625^2) = 0.1(0.05 + 1.11857) = 0.11686
k_4 = h f(0 + 0.1, 1 + 0.11686) = 0.1(0.1 + 1.11686^2) = 0.1(0.1 + 1.24738) = 0.13474
Δ y_0 = (1 / 6)(k_1 + 2k_2 + 2k_3 + k_4) = (1 / 6)(0.1 + 0.2305 + 0.23372 + 0.13474) = (0.69896 / 6) = 0.11649
Thus, y(0.1) = 1 + 0.11649 = **1.11649**.

STEP 2: From x_1 = 0.1, y_1 = 1.11649 to x_2 = 0.2:
k_1 = 0.1 [0.1 + (1.11649)^2] = 0.1 [0.1 + 1.24655] = 0.13466
k_2 = 0.1 [0.15 + (1.11649 + 0.06733)^2] = 0.1 [0.15 + (1.18382)^2] = 0.1 [0.15 + 1.40143] = 0.15514
k_3 = 0.1 [0.15 + (1.11649 + 0.07757)^2] = 0.1 [0.15 + (1.19406)^2] = 0.1 [0.15 + 1.42578] = 0.15758
k_4 = 0.1 [0.2 + (1.11649 + 0.15758)^2] = 0.1 [0.2 + (1.27407)^2] = 0.1 [0.2 + 1.62325] = 0.18233
Δ y_1 = (1 / 6)(0.13466 + 2(0.15514) + 2(0.15758) + 0.18233) = (0.94243 / 6) = 0.15707
Thus, y(0.2) = 1.11649 + 0.15707 = **1.27356**.
Final Answer: y(0.2) ≈ **1.2736**.

Q2 Marks: 15M / 3M / 7.5M Repeated 7x CCSU: 2024, 2023, 2022, 2021, 2020, 2019, 2018

Obtain a real root of the equation f(x) = x^3 + x^2 - 1 = 0 using Bisection method correct to three decimal places.

Solution: Given f(x) = x^3 + x^2 - 1 = 0.
Evaluating at test points: f(0) = -1 < 0, f(1) = 1 + 1 - 1 = +1 > 0.
Since f(0) · f(1) < 0, a real root lies in [0, 1].
Iteration 1: x_1 = (0 + 1 / 2) = 0.5 ⇒ f(0.5) = 0.125 + 0.25 - 1 = -0.625 < 0 ⇒ root ∈ [0.5, 1.0].
Iteration 2: x_2 = (0.5 + 1.0 / 2) = 0.75 ⇒ f(0.75) = 0.421875 + 0.5625 - 1 = -0.015625 < 0 ⇒ root ∈ [0.75, 1.0].
Iteration 3: x_3 = (0.75 + 1.0 / 2) = 0.875 ⇒ f(0.875) = 0.6699 + 0.7656 - 1 = +0.4355 > 0 ⇒ root ∈ [0.75, 0.875].
Iteration 4: x_4 = (0.75 + 0.875 / 2) = 0.8125 ⇒ f(0.8125) = +0.1963 > 0 ⇒ root ∈ [0.75, 0.8125].
Iteration 5: x_5 = (0.75 + 0.8125 / 2) = 0.78125 ⇒ f(0.78125) = +0.0872 > 0 ⇒ root ∈ [0.75, 0.78125].
Iteration 6: x_6 = (0.75 + 0.78125 / 2) = 0.765625 ⇒ f(0.7656) = +0.0349 > 0 ⇒ root ∈ [0.75, 0.765625].
Iteration 7: x_7 = (0.75 + 0.765625 / 2) = 0.7578 ⇒ f(0.7578) = +0.0094 > 0 ⇒ root ∈ [0.75, 0.7578].
Iteration 8: x_8 = (0.75 + 0.7578 / 2) = 0.7539 ⇒ f(0.7539) = -0.0031 < 0 ⇒ root ∈ [0.7539, 0.7578].
Iteration 9: x_9 = 0.7558.
Root correct to 3 decimal places: x ≈ 0.755.

Q3 **Marks: 15M / 3M / 7.5M** **Repeated 7x** **CCSU: 2024, 2023, 2022, 2021, 2020, 2019, 2018**

Solve the differential equation $dy/dx = x + y^2$, $y(0) = 1$ using Picard's method (two approximations).

Solution: Given y' = x + y^2 with initial condition x_0 = 0, y_0 = 1.
Picard's iterative formula: y^{(n)}(x) = y_0 + ∫_{x_0}^x f(t, y^{(n-1)}(t)) \, dt.
First Approximation (n = 1):
y^{(1)}(x) = 1 + ∫_0^x (t + 1^2) \, dt = 1 + [(t^2 / 2) + t]_0^x = **1 + x + \frac{x^2}{2}**
Second Approximation (n = 2):
y^{(2)}(x) = 1 + ∫_0^x [t + (1 + t + (t^2 / 2))^2] dt = 1 + ∫_0^x [t + 1 + 2t + 2t^2 + t^3 + (t^4 / 4)] dt
= 1 + ∫_0^x (1 + 3t + 2t^2 + t^3 + (t^4 / 4)) dt = **1 + x + \frac{3}{2}x^2 + (2 / 3)x^3 + (x^4 / 4) + (x^5 / 20)**

Q4 **Marks: 15M / 22.5M / 3M / 7.5M** **Repeated 6x** **CCSU: 2024, 2023, 2022, 2021, 2019, 2018**

Evaluate $\int_0^1 dx/(1 + x^2)$ using Simpson's 3/8 rule. Hence obtain the approximate value of π.

Solution: Let f(x) = (1 / 1 + x^2) over interval [0, 1].
For Simpson's 3/8 rule, number of subintervals n must be a multiple of 3. Choose n = 6, so h = (1 - 0 / 6) = (1 / 6).

i	x_i	y_i = 1 / (1 + x_i^2)	Decimal (5 places)
0	0	1 / 1	1.00000
1	1/6	36 / 37	0.97297
2	2/6 = 1/3	9 / 10	0.90000
3	3/6 = 1/2	4 / 5	0.80000
4	4/6 = 2/3	9 / 13	0.69231
5	5/6	36 / 61	0.59016
6	1	1 / 2	0.50000

Simpson's 3/8 formula:
I = (3h / 8) [(y_0 + y_6) + 3(y_1 + y_2 + y_4 + y_5) + 2(y_3)]
Calculate components: y_0 + y_6 = 1.00000 + 0.50000 = 1.50000
y_1 + y_2 + y_4 + y_5 = 0.97297 + 0.90000 + 0.69231 + 0.59016 = 3.15544
3 × 3.15544 = 9.46632
2 × y_3 = 2 × 0.80000 = 1.60000
Sum inside brackets = 1.50000 + 9.46632 + 1.60000 = 12.56632.
I = (3 × (1/6) / 8) × 12.56632 = (1 / 16) × 12.56632 = **0.785395**
Since ∫_0^1 (dx / 1 + x^2) = [\tan^{-1}x]_0^1 = (π / 4):
π ≈ 4 × I = 4 × 0.785395 = **3.14158**
Final Answer: I ≈ 0.78540 and π ≈ 3.1416.

Q5 **Marks: 15M / 3M** **Repeated 6x** **CCSU: 2024, 2023, 2022, 2020, 2019, 2018**

Solve the system of linear equations by Gauss-Seidel Iteration method: 6x + y + z = 20, x + 4y - z = 6, x - y + 5z = 7.

Solution: The system is diagonally dominant because |6| > |1|+|1|, |4| > |1|+|-1|, |5| > |1|+|-1|.
Rewrite in iterative form:
x = (20 - y - z / 6), \quad y = (6 - x + z / 4), \quad z = (7 - x + y / 5)
Start with initial guess x^{(0)} = 0, y^{(0)} = 0, z^{(0)} = 0:

Iteration 1: x^{(1)} = (20 - 0 - 0 / 6) = 3.3333
y^{(1)} = (6 - 3.3333 + 0 / 4) = 0.6667

z^{(1)} = (7 - 3.3333 + 0.6667 / 5) = 0.8667

Iteration 2: x^{(2)} = (20 - 0.6667 - 0.8667 / 6) = (18.4666 / 6) = 3.0778
y^{(2)} = (6 - 3.0778 + 0.8667 / 4) = (3.7889 / 4) = 0.9472
z^{(2)} = (7 - 3.0778 + 0.9472 / 5) = (4.8694 / 5) = 0.9739

Iteration 3: x^{(3)} = (20 - 0.9472 - 0.9739 / 6) = (18.0789 / 6) = 3.0132
y^{(3)} = (6 - 3.0132 + 0.9739 / 4) = (3.9607 / 4) = 0.9902
z^{(3)} = (7 - 3.0132 + 0.9902 / 5) = (4.9770 / 5) = 0.9954

Iteration 4: x^{(4)} = (20 - 0.9902 - 0.9954 / 6) = 3.0024
y^{(4)} = (6 - 3.0024 + 0.9954 / 4) = 0.9983
z^{(4)} = (7 - 3.0024 + 0.9983 / 5) = 0.9992

Final Converged Solution: x ≈ 3.000, \; y ≈ 1.000, \; z ≈ 1.000.

Q6 **Marks: 22.5M / 3M / 7.5M** 🔥 **Repeated 5x** **CCSU: 2024, 2023, 2022, 2021, 2020**

Using method of False Position (Regula-Falsi), find a positive root of the equation $x^3 - 2x - 5 = 0$.

Solution: Let f(x) = x^3 - 2x - 5.
f(2) = 8 - 4 - 5 = -1 < 0
f(3) = 27 - 6 - 5 = +16 > 0.
Root lies between a = 2 and b = 3.
Iter 1: x_1 = (a f(b) - b f(a) / f(b) - f(a)) = (2(16) - 3(-1) / 16 - (-1)) = (32 + 3 / 17) = (35 / 17) = 2.0588.
f(2.0588) = (2.0588)^3 - 2(2.0588) - 5 = 8.7228 - 4.1176 - 5 = -0.3948 < 0.
Root lies between 2.0588 and 3.

Iter 2: x_2 = (2.0588(16) - 3(-0.3948) / 16 - (-0.3948)) = (32.9408 + 1.1844 / 16.3948) = (34.1252 / 16.3948) = 2.0813.
f(2.0813) = 9.0125 - 4.1626 - 5 = -0.1501 < 0.

Iter 3: x_3 = (2.0813(16) - 3(-0.1501) / 16 + 0.1501) = (33.3008 + 0.4503 / 16.1501) = (33.7511 / 16.1501) = **2.0898**.
f(2.0898) ≈ -0.05.
Iter 4: x_4 = **2.0945** ≈ **2.094**.
Final Root: x ≈ 2.094.

Q7 **Marks: 15M / 3M** ★ **Repeated 3x** **CCSU: 2024, 2022, 2021**

Find the value of dy/dx at x = 1.2 for the following ordered pairs: (1.0, 2.7183), (1.2, 3.3201), (1.4, 4.0552), (1.6, 4.9530), (1.8, 6.0496), (2.0, 7.3891), (2.2, 9.0250).

Solution: Here step size h = 0.2. We evaluate the derivative at x_0 = 1.2 (using forward differences with base x_0 = 1.2):

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$
1.2	3.3201	0.7351	0.1627	0.0339
1.4	4.0552	0.8978	0.1966	0.0429
1.6	4.9530	1.0966	0.2429	-
1.8	6.0496	1.3395	-	-
2.0	7.3891	-	-	-

Using Newton's forward difference formula for derivative at the first point (u=0):
((dy / dx))_{x=1.2} = (1 / h) [Δ y_0 - (1 / 2)Δ^2 y_0 + (1 / 3)Δ^3 y_0]

= (1 / 0.2) [0.7351 - (0.1627 / 2) + (0.0339 / 3)] = 5 × [0.7351 - 0.08135 + 0.01130] = 5 × 0.66505 = **3.3253**.
Note: Analytical derivative of y = e^x at 1.2 is e^{1.2} = 3.3201; numerical formula gives **3.325**.

Q8 **Marks: 15M** ★ Repeated 3x **CCSU: 2023, 2022, 2019**

Using Lagrange's interpolation formula, find the value of y corresponding to x = 2 from: (1, 1), (3, 27), (5, 125), (6, 216).

Solution: Points: (x_0, y_0) = (1, 1), (x_1, y_1) = (3, 27), (x_2, y_2) = (5, 125), (x_3, y_3) = (6, 216) (Note: y = x^3).
Evaluating at x = 2 by Lagrange formula:
y(2) = ∑_{i=0}^3 L_i(2) y_i
L_0(2) = ((2-3)(2-5)(2-6) / (1-3)(1-5)(1-6)) = ((-1)(-3)(-4) / (-2)(-4)(-5)) = (-12 / -40) = 0.300
L_1(2) = ((2-1)(2-5)(2-6) / (3-1)(3-5)(3-6)) = ((1)(-3)(-4) / (2)(-2)(-3)) = (12 / 12) = 1.000
L_2(2) = ((2-1)(2-3)(2-6) / (5-1)(5-3)(5-6)) = ((1)(-1)(-4) / (4)(2)(-1)) = (4 / -8) = -0.500
L_3(2) = ((2-1)(2-3)(2-5) / (6-1)(6-3)(6-5)) = ((1)(-1)(-3) / (5)(3)(1)) = (3 / 15) = 0.200

Sum of weights = 0.3 + 1.0 - 0.5 + 0.2 = 1.0.
Compute y(2):
y(2) = 0.3(1) + 1.0(27) - 0.5(125) + 0.2(216) = 0.3 + 27 - 62.5 + 43.2 = **8.0**
Answer: y(2) = 8 (Exact value of 2^3).

Q9 **Marks: 15M / 7.5M** ★ Repeated 3x **CCSU: 2021, 2020, 2019**

Write the conditions for applicability of: (a) Trapezoidal rule, (b) Simpson's 1/3 rule. Evaluate $\int_0^1 dx/(1+x)$ using both rules with h = 0.25.

Solution:
Conditions:
(a) *Trapezoidal Rule:* The number of subintervals n can be any positive integer (even or odd).
(b) *Simpson's 1/3 Rule:* The number of subintervals n must be even (n = 2m).

Evaluation for $\int_0^1 dx/(1+x)$, h = 0.25:
y_0 = 1.0, y_1 = 0.8, y_2 = 0.6667, y_3 = 0.5714, y_4 = 0.5.
- *Trapezoidal:* I_T = (0.25 / 2) [1.5 + 2(0.8 + 0.6667 + 0.5714)] = 0.125 × 5.5762 = **0.6970**.
- *Simpson's 1/3:* I_S = (0.25 / 3) [1.5 + 4(0.8 + 0.5714) + 2(0.6667)] = (0.25 / 3)[1.5 + 5.4856 + 1.3334] = (0.25 / 3)(8.319) = **0.6932**.
Exact value: \ln(2) ≈ 0.69315.

Q10 **Marks: 15M** ★ Repeated 2x **CCSU: 2022, 2020**

Solve the system by Gauss-Seidel iterative method: 20x + y - 2z = 17, 2x + 20y - 2z = -18, 2x - 3y + 20z = 25.

Solution: The system is strictly diagonally dominant:
x = (17 - y + 2z / 20), \quad y = (-18 - 2x + 2z / 20), \quad z = (25 - 2x + 3y / 20)
Start with (x_0, y_0, z_0) = (0, 0, 0).
Iter 1: x_1 = (17 / 20) = 0.8500
y_1 = (-18 - 2(0.85) / 20) = (-19.7 / 20) = -0.9850
z_1 = (25 - 2(0.85) + 3(-0.985) / 20) = (25 - 1.7 - 2.955 / 20) = 1.0173

Iter 2: x_2 = (17 - (-0.9850) + 2(1.0173) / 20) = (20.0196 / 20) = 1.0010
y_2 = (-18 - 2(1.0010) + 2(1.0173) / 20) = (-17.9674 / 20) = -0.9984
z_2 = (25 - 2(1.0010) + 3(-0.9984) / 20) = (20.0028 / 20) = 1.0001

Iter 3: x_3 = (17 - (-0.9984) + 2(1.0001) / 20) = 1.0000
y_3 = (-18 - 2(1) + 2(1) / 20) = -1.0000
z_3 = (25 - 2(1) + 3(-1) / 20) = 1.0000
Final Answer: x = 1.000, \; y = -1.000, \; z = 1.000.

Q11 **Marks: 22.5M** CCSU Exam **CCSU: 2021**

Interpolate by means of Gauss's formula the population for the year 1936, given: [1901: 12k, 1911: 15k, 1921: 20k, 1931: 27k, 1941: 39k, 1951: 52k].

Solution: Origin at x_0 = 1931, h = 10 \implies u = (1936 - 1931 / 10) = 0.5.
Central difference table around 1931 (y_0 = 27):
Δ y_0 = 39 - 27 = 12, Δ y_{-1} = 27 - 20 = 7, Δ^2 y_{-1} = 12 - 7 = 5.
Gauss Forward Interpolation formula:
y = y_0 + u Δ y_0 + (u(u-1) / 2) Δ^2 y_{-1} + ((u+1)u(u-1) / 6) Δ^3 y_{-1}
y(1936) = 27 + 0.5(12) + (0.5(-0.5) / 2)(5) = 27 + 6 - 0.625 = **32.375** thousand.
Population in 1936: ≈ 32,375.

Q12 **Marks: 22.5M** CCSU Exam **CCSU: 2021**

Find the solution of the system by Gauss-Seidel Method: 27x + 6y - z = 85, 6x + 15y + 2z = 72, x + y + 54z = 110.

Solution: Diagonally dominant formulas:
x = (85 - 6y + z / 27), \quad y = (72 - 6x - 2z / 15), \quad z = (110 - x - y / 54)
Start (x_0, y_0, z_0) = (0, 0, 0).
Iter 1: x_1 = 3.1481, y_1 = 3.5407, z_1 = 1.9132.
Iter 2: x_2 = 2.4322, y_2 = 3.5720, z_2 = 1.9258.
Iter 3: x_3 = 2.4257, y_3 = 3.5730, z_3 = 1.9260.
Converged solution: **x ≈ 2.426, \; y ≈ 3.573, \; z ≈ 1.926**.

Q13 **Marks: 15M** CCSU Exam **CCSU: 2024**

Find the Lagrange interpolating polynomial of degree 2 approximating y = $\log_e(x)$ defined by values: (2.0, 0.69315), (2.5, 0.91629), (3.0, 1.09861). Hence determine the value of $\log_e(2.7)$.

Solution: Data points: x_0 = 2.0, y_0 = 0.69315; x_1 = 2.5, y_1 = 0.91629; x_2 = 3.0, y_2 = 1.09861.
Lagrange interpolation formula for 3 points:
P_2(x) = ((x - x_1)(x - x_2) / (x_0 - x_1)(x_0 - x_2)) y_0 + ((x - x_0)(x - x_2) / (x_1 - x_0)(x_1 - x_2)) y_1 + ((x - x_0)(x - x_1) / (x_2 - x_0)(x_2 - x_1)) y_2
For x = 2.7: L_0(2.7) = ((2.7 - 2.5)(2.7 - 3.0) / (2.0 - 2.5)(2.0 - 3.0)) = ((0.2)(-0.3) / (-0.5)(-1.0)) = (-0.06 / 0.5) = -0.12
L_1(2.7) = ((2.7 - 2.0)(2.7 - 3.0) / (2.5 - 2.0)(2.5 - 3.0)) = ((0.7)(-0.3) / (0.5)(-0.5)) = (-0.21 / -0.25) = +0.84
L_2(2.7) = ((2.7 - 2.0)(2.7 - 2.5) / (3.0 - 2.0)(3.0 - 2.5)) = ((0.7)(0.2) / (1.0)(0.5)) = (0.14 / 0.5) = +0.28

Check sum of weights: -0.12 + 0.84 + 0.28 = 1.00 (exact).
Now compute P_2(2.7):
P_2(2.7) = -0.12(0.69315) + 0.84(0.91629) + 0.28(1.09861)
= -0.083178 + 0.769684 + 0.307611 = **0.993245** ≈ **0.9932**.
Verification: Exact \ln(2.7) ≈ 0.9932518.

Q14 **Marks: 15M** CCSU Exam **CCSU: 2021**

By means of Newton's divided difference formula find f(8) and f(15) from: x = [4, 5, 7, 10, 11, 13], f(x) = [48, 100, 294, 900, 1210, 2028].

Solution: Construct divided difference table:

x	f(x)	1st DD	2nd DD	3rd DD	4th DD
4	48	52	15	1	0
5	100	97	21	1	-
7	294	202	27	-	-
10	900	310	-	-	-
11	1210	409	-	-	-
13	2028	-	-	-	-

Newton's Divided Difference formula:
f(x) = 48 + 52(x - 4) + 15(x - 4)(x - 5) + 1(x - 4)(x - 5)(x - 7)
For x = 8: f(8) = 48 + 52(4) + 15(4)(3) + 1(4)(3)(1) = 48 + 208 + 180 + 12 = **448**.
For x = 15: f(15) = 48 + 52(11) + 15(11)(10) + 1(11)(10)(8) = 48 + 572 + 1650 + 880 = **3150**.
Answer: f(8) = 448, f(15) = 3150.

Q15 **Marks: 15M** CCSU Exam **CCSU: 2020**

Apply Lagrange's formula to find the cubic polynomial which includes: (0, 2), (1, 3), (2, 12), (5, 147).

Solution: Applying Lagrange interpolation across x = [0, 1, 2, 5], y = [2, 3, 12, 147]:
Notice y = x^3 + 2:
x=0 \implies y=2;
x=1 \implies y=3;
x=2 \implies y=8+2=10 ≈ 12.
Computing Lagrange polynomial:
y(x) = x^3 + x^2 - x + 2
Verification: y(0)=2, y(1)=3, y(2)=8+4-2+2=12, y(5)=125+25-5+2=147. (Exact!)

Q16 **Marks: 15M** CCSU Exam **CCSU: 2018**

Using Stirling formula, find f(28) from table: f(20)=49225, f(25)=48316, f(30)=47236, f(35)=45926, f(40)=44306.

Solution: Equidistant table with h = 5. We need f(28).
Choose origin x_0 = 30 (closest center point). Then u = (28 - 30 / 5) = -0.4.
Central difference table:

u	x	y = f(x)	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
-2	20	49225	-909			
-1	25	48316	-1080	-171	-59	
0	30	47236	-1310	-230	-90	-31
1	35	45926	-1620	-310		
2	40	44306				

Stirling's Formula:
f(x) = y_0 + u \frac{Δ y_{-1} + Δ y_0}{2} + (u^2 / 2)Δ^2 y_{-1} + (u(u^2 - 1) / 6)\frac{Δ^3 y_{-2} + Δ^3 y_{-1}}{2} + (u^2(u^2 - 1) / 24)Δ^4 y_{-2}
Substituting values: y_0 = 47236
\frac{Δ y_{-1} + Δ y_0}{2} = (-1080 + (-1310) / 2) = -1195
u × (-1195) = (-0.4) × (-1195) = +478.0
(u^2 / 2)Δ^2 y_{-1} = (0.16 / 2) × (-230) = -18.4
(u(u^2 - 1) / 6) \frac{Δ^3 y_{-2} + Δ^3 y_{-1}}{2} = (-0.4(0.16 - 1) / 6) × (-59 - 90 / 2) = (0.336 / 6) × (-74.5) = -4.172
(u^2(u^2 - 1) / 24)Δ^4 y_{-2} = (0.16(-0.84) / 24) × (-31) = +0.174
Summing terms:
f(28) = 47236 + 478.0 - 18.4 - 4.172 + 0.174 = **47691.6 ≈ 47692**
Answer: f(28) ≈ 47692.

SECTION B • MEDIUM ANSWER QUESTIONS (7.5 MARKS / CORE CONCEPTUAL SOLUTIONS)

9 Questions

Q17 Marks: 3M / 7.5M ★ Repeated 3x CCSU: 2021, 2019, 2018

Solve by Gauss elimination: 8x - 3y + 2z = 23, 4x + 11y - z = 33, 6x + 3y + 12z = 36.

Solution: Augmented matrix: \begin{bmatrix} 8 & -3 & 2 & | & 23 \\ 4 & 11 & -1 & | & 33 \\ 6 & 3 & 12 & | & 36 \end{bmatrix}
Divide row 3 by: \begin{bmatrix} 2 & 1 & 4 & | & 12 \end{bmatrix}. Interchange R_1 ≤ftrightarrow R_3:
\begin{bmatrix} 2 & 1 & 4 & | & 12 \\ 4 & 11 & -1 & | & 33 \\ 8 & -3 & 2 & | & 23 \end{bmatrix}
R_2 → R_2 - 2R_1: [0, 9, -9 | 9] \implies y - z = 1.
R_3 → R_3 - 4R_1: [0, -7, -14 | -25].
Substitute y = 1 + z into row 3: -7(1 + z) - 14z = -25 \implies -7 - 21z = -25 \implies -21z = -18 \implies **z = \frac{6}{7}** ≈ 0.857}.
y = 1 + \frac{6}{7} = (13 / 7) ≈ 1.857}.
2x + y + 4z = 12 \implies 2x = 12 - (13 / 7) - (24 / 7) = 12 - (37 / 7) = (47 / 7) \implies **x = \frac{47}{14}** ≈ 3.357}.
Answer: x ≈ 3.357, \; y ≈ 1.857, \; z ≈ 0.857.

Q18 Marks: 7.5M ★ Repeated 2x CCSU: 2024, 2022

Find the missing terms in the following table:
x: 0, 5, 10, 15, 20, 25
y: 1, 3, y_2, 73, 225, y_5

Solution: We are given 4 known values of y (y_0=1, y_1=3, y_3=73, y_4=225) out of 6 points. Therefore, the function can be approximated by a polynomial of degree 3, which implies fourth differences vanish:
Δ^4 y_0 = 0 \quad and \quad Δ^4 y_1 = 0
Expressing in terms of shift operator E = 1 + Δ:
(E - 1)^4 y_0 = y_4 - 4y_3 + 6y_2 - 4y_1 + y_0 = 0

Substitute known values: 225 - 4(73) + 6y_2 - 4(3) + 1 = 0 \implies 225 - 292 + 6y_2 - 12 + 1 = 0
6y_2 - 78 = 0 \implies **y_2 = 13**.

Next, for y_5:
(E - 1)^4 y_1 = y_5 - 4y_4 + 6y_3 - 4y_2 + y_1 = 0
Substitute: y_5 - 4(225) + 6(73) - 4(13) + 3 = 0
y_5 - 900 + 438 - 52 + 3 = 0 \implies y_5 - 511 = 0 \implies **y_5 = 511**.
Answer: The missing terms are **y_2 = 13** and **y_5 = 511**.

Q19 **Marks: 7.5M** ★ Repeated 2x **CCSU: 2023, 2019**

Using Newton's forward interpolation formula, find the cubic polynomial which takes the values: x = [0, 1, 2, 3], y = [1, 2, 13, 40].

Solution: Here x_0 = 0, h = 1. Form the forward difference table:

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$
0	1	1	10	6
1	2	11	16	-
2	13	27	-	-
3	40	-	-	-

Here u = (x - 0 / 1) = x.
Newton's forward interpolation formula:
y(x) = y_0 + u Δ y_0 + (u(u-1) / 2!) Δ^2 y_0 + (u(u-1)(u-2) / 3!) Δ^3 y_0
Substitute values: y(x) = 1 + x(1) + (x(x-1) / 2)(10) + (x(x-1)(x-2) / 6)(6)
= 1 + x + 5(x^2 - x) + (x^3 - 3x^2 + 2x)
= 1 + x + 5x^2 - 5x + x^3 - 3x^2 + 2x
y(x) = x^3 + 2x^2 - 2x + 1
Verification: y(0)=1, y(1)=1+2-2+1=2, y(2)=8+8-4+1=13, y(3)=27+18-6+1=40. (Exact!)

Q20 **Marks: 3M / 7.5M** ★ Repeated 2x **CCSU: 2021, 2020**

Use Euler's Method with h = 0.1 to find the solution of $dy/dx = x + y^2$, y(0) = 0 in range 0 ≤ x ≤ 0.5.

Solution: y_{n+1} = y_n + h f(x_n, y_n) = y_n + 0.1(x_n + y_n^2).
1. x_0 = 0.0, y_0 = 0 \implies y_1 = 0 + 0.1(0) = **0.0000** at x=0.1.
2. x_1 = 0.1, y_1 = 0 \implies y_2 = 0 + 0.1(0.1 + 0) = **0.0100** at x=0.2.
3. x_2 = 0.2, y_2 = 0.01 \implies y_3 = 0.01 + 0.1(0.2 + 0.0001) = **0.0300** at x=0.3.
4. x_3 = 0.3, y_3 = 0.03 \implies y_4 = 0.03 + 0.1(0.3 + 0.0009) = **0.0601** at x=0.4.
5. x_4 = 0.4, y_4 = 0.0601 \implies y_5 = 0.0601 + 0.1(0.4 + 0.0036) = **0.1005** at x=0.5.
Answer: y(0.1)=0, y(0.2)=0.01, y(0.3)=0.03, y(0.4)=0.0601, y(0.5)=0.1005.

Q21 **Marks: 7.5M** CCSU Exam **CCSU: 2021**

Find the form of the function from following given data: x = [1, 2, 3, 4], f(x) = [3, 6, 11, 18].

Solution: Step h = 1, x_0 = 1. Form difference table:

x	f(x)	Δf	$\Delta^2 f$	$\Delta^3 f$
1	3	3	2	0
2	6	5	2	-
3	11	7	-	-
4	18	-	-	-

Here Δ^2 f = 2 is constant, and Δ^3 f = 0. Let u = x - 1:
f(x) = f_0 + u Δ f_0 + (u(u-1) / 2) Δ^2 f_0 = 3 + (x-1)(3) + ((x-1)(x-2) / 2)(2)
= 3 + 3x - 3 + (x^2 - 3x + 2) = **x^2 + 2**.
Answer: The function is **f(x) = x^2 + 2**.

Q22 **Marks: 7.5M** CCSU Exam **CCSU: 2020**

Find backward differences for the points: (5, 11), (6, 13), (9, 28), (11, 40).

Solution: Since step size is unequal, using divided differences: f[5, 6] = (13 - 11 / 6 - 5) = 2.
f[6, 9] = (28 - 13 / 9 - 6) = (15 / 3) = 5.
f[9, 11] = (40 - 28 / 11 - 9) = (12 / 2) = 6.
Second divided differences: f[5, 6, 9] = (5 - 2 / 9 - 5) = (3 / 4) = 0.75.
f[6, 9, 11] = (6 - 5 / 11 - 6) = (1 / 5) = 0.20.

Q23 **Marks: 7.5M** CCSU Exam **CCSU: 2020**

Using Bessel's formula, find y(25) from: x = [20, 24, 28, 32], y = [24, 32, 35, 40].

Solution: h = 4. Choose origin x_0 = 24. Then u = (25 - 24 / 4) = 0.25.
y_{-1} = 24, y_0 = 32, y_1 = 35, y_2 = 40.
Δ y_0 = 35 - 32 = 3, Δ y_{-1} = 32 - 24 = 8, Δ y_1 = 40 - 35 = 5.
Δ^2 y_{-1} = 3 - 8 = -5, Δ^2 y_0 = 5 - 3 = 2.
Bessel's interpolation formula:
y = (y_0 + y_1 / 2) + (u - (1 / 2))Δ y_0 + (u(u-1) / 4)(Δ^2 y_{-1} + Δ^2 y_0)
(32 + 35 / 2) + (0.25 - 0.5)(3) + (0.25(-0.75) / 4)(-5 + 2)
= 33.5 - 0.75 + (-0.1875 / 4)(-3) = 32.75 + 0.1406 = **32.89**.

Q24 **Marks: 7.5M** CCSU Exam **CCSU: 2018**

Given f(0) = 16.35, f(5) = 14.88, f(10) = 13.59, f(15) = 12.46. Find x when f(x) = 14.00 by Inverse Interpolation.

Solution: Using **Lagrange's Inverse Interpolation Formula** where y is independent and x is dependent:

i	x_i	y_i = f(x_i)	y - y_i (for y = 14.00)
0	0	16.35	-2.35
1	5	14.88	-0.88
2	10	13.59	+0.41
3	15	12.46	+1.54

Lagrange inverse interpolation: x = ∑_{i=0}^3 x_i \prod_{j ≠q i} (y - y_j / y_i - y_j).
Term for i=0 is 0 since x_0 = 0.
Term 1 (i=1): 5 × ((14-16.35)(14-13.59)(14-12.46) / (14.88-16.35)(14.88-13.59)(14.88-12.46)) = 5 × ((-2.35)(0.41)(1.54) / (-1.47)

(1.29)(2.42)) = 5 × (-1.4838 / -4.5888) = 5 × 0.3233 = 1.6166
Term 2 (i=2): 10 × ((14-16.35)(14-14.88)(14-12.46) / (13.59-16.35)(13.59-14.88)(13.59-12.46)) = 10 × ((-2.35)(-0.88)(1.54) / (-2.76)(-1.29)(1.13)) = 10 × (3.1847 / 4.0233) = 10 × 0.7916 = 7.9156
Term 3 (i=3): 15 × ((14-16.35)(14-14.88)(14-13.59) / (12.46-16.35)(12.46-14.88)(12.46-13.59)) = 15 × ((-2.35)(-0.88)(0.41) / (-3.89)(-2.42)(-1.13)) = 15 × (0.8479 / -10.637) = -1.1957
Summing terms: x = 0 + 1.6166 + 7.9156 - 1.1957 = **8.3365**.
Answer: For f(x) = 14.00, x ≈ 8.34.

Q25 **Marks: 7.5M** CCSU Exam **CCSU: 2018**

Estimate the sales for 1966 using the data: Years = [1931, 1941, 1951, 1961, 1971, 1981], Sales = [12, 15, 20, 27, 39, 52].

Solution: Equidistant data with h = 10 years. We estimate for x = 1966.
Take origin x_0 = 1961 (center). Then u = (1966 - 1961 / 10) = 0.5.
Difference table for sales y:

Year	Sales (y)	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
1931	12	3	2	0	3
1941	15	5	2	3	-2
1951	20	7	5	1	
1961	27	12	6		
1971	39	13			
1981	52				

Using **Newton's Forward Interpolation** from x_0 = 1961, y_0 = 27, u = 0.5:
y(1966) = y_0 + uΔ y_0 + (u(u-1) / 2)Δ^2 y_0 = 27 + (0.5)(12) + (0.5(-0.5) / 2)(1) = 27 + 6 - 0.125 = **32.875** thousand
Estimated Sales for 1966: Approximately **32,875 units**.

SECTION A • SHORT ANSWER QUESTIONS (3 MARKS / PRECISE DEFINITIVE SOLUTIONS)

8 Questions

Q26 **Marks: 3M** ★ Repeated 2x **CCSU: 2024, 2022**

Define backward differences. Make backward difference table for the ordered pairs: (x_0, y_0), (x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4).

Definition: Let y = f(x) be tabulated at equidistant points x_0, x_1, \dots, x_n with step h. The first backward difference operator ∇ is defined as:
∇ y_k = y_k - y_{k-1} \quad (k = 1, 2, \dots, n)
Higher-order backward differences are given recursively by ∇^r y_k = ∇^{r-1} y_k - ∇^{r-1} y_{k-1}.
Backward Difference Table:

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	∇&sup4;y
x_0	y_0	-	-	-	-
x_1	y_1	$\nabla y_1 = y_1 - y_0$	-	-	-
x_2	y_2	$\nabla y_2 = y_2 - y_1$	$\nabla^2 y_2 = \nabla y_2 - \nabla y_1$	-	-
x_3	y_3	$\nabla y_3 = y_3 - y_2$	$\nabla^2 y_3 = \nabla y_3 - \nabla y_2$	$\nabla^3 y_3 = \nabla^2 y_3 - \nabla^2 y_2$	-
x_4	y_4	$\nabla y_4 = y_4 - y_3$	$\nabla^2 y_4 = \nabla y_4 - \nabla y_3$	$\nabla^3 y_4 = \nabla^2 y_4 - \nabla^2 y_3$	∇&sup4;$y_4 = \nabla^3 y_4 - \nabla^3 y_3$

Q27 **Marks: 3M** **★ Repeated 2x** **CCSU: 2023, 2018**

Find √12 by applying Newton-Raphson's method.

Solution: Let $x = \sqrt{12} \implies x^2 - 12 = 0$.
Define $f(x) = x^2 - 12$ and $f'(x) = 2x$.
Newton-Raphson iterative formula:
x_{n+1} = x_n - (f(x_n) / f'(x_n)) = x_n - (x_n^2 - 12 / 2x_n) = (1 / 2)(x_n + (12 / x_n))
Since $3^2 = 9 < 12 < 4^2 = 16$, take initial guess x_0 = 3.5.
Iteration 1: x_1 = 0.5 × (3.5 + 12/3.5) = 0.5 × (3.5 + 3.42857) = 3.46429
Iteration 2: x_2 = 0.5 × (3.46429 + 12/3.46429) = 0.5 × (3.46429 + 3.46391) = 3.46410
Iteration 3: x_3 = 0.5 × (3.46410 + 12/3.46410) = 3.46410
Final Answer: √12 ≈ 3.46410 (correct to 5 decimal places).

Q28 **Marks: 3M** **★ Repeated 2x** **CCSU: 2020, 2019**

Define operators E, Δ and ∇. Also obtain relation between E and Δ.

Definitions:
1. **Shift Operator E:** E f(x) = f(x + h).
2. **Forward Difference Δ:** Δ f(x) = f(x + h) - f(x).
3. **Backward Difference ∇:** ∇ f(x) = f(x) - f(x - h).

Relation between E and Δ:
Δ f(x) = f(x + h) - f(x) = E f(x) - f(x) = (E - 1) f(x) \implies **Δ = E - 1 \iff E = 1 + Δ**.

Q29 **Marks: 3M** **★ Repeated 2x** **CCSU: 2019, 2018**

Prepare a divided difference table for the given data: x = [1, 2, 4, 7, 12], f(x) = [22, 30, 82, 106, 216].

Solution: Divided differences are calculated as f[x_i, x_{i+1}] = \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i}:

x	f(x)	1st Div. Diff	2nd Div. Diff	3rd Div. Diff	4th Div. Diff
1	22	(30-22)/(2-1) = **8**			
2	30	(82-30)/(4-2) = **26**	(26-8)/(4-1) = **6**		
4	82	(106-82)/(7-4) = **8**	(8-26)/(7-2) = **-3.6**	(-3.6 - 6)/(7-1) = **-1.6**	
7	106	(216-106)/(12-7) = **22**	(22-8)/(12-4) = **1.75**	(1.75 - (-3.6))/(12-2) = **0.535**	(0.535 - (-1.6))/(12-1) = **0.194**
12	216				

Q30 **Marks: 3M** CCSU Exam **CCSU: 2023**

Prove that E ∇ = ∇ E = Δ, where E is shift operator, ∇ is backward difference operator, and Δ is forward difference operator.

Proof:
1. *Part 1: Prove E ∇ = Δ*
By definition, ∇ f(x) = f(x) - f(x-h) = (1 - E^{-1}) f(x) \implies ∇ = 1 - E^{-1}.
Therefore,
E ∇ = E (1 - E^{-1}) = E - E · E^{-1} = E - 1
Since forward difference operator is Δ = E - 1, we obtain:
E ∇ = Δ
2. *Part 2: Prove ∇ E = Δ*
∇ E = (1 - E^{-1}) E = E - E^{-1} E = E - 1 = **Δ**
Hence, **E ∇ = ∇ E = Δ**. (Hence Proved).

Q31 **Marks: 3M** CCSU Exam **CCSU: 2023**

Find the first derivative of the function at x = 1.2 given: x = [1, 2, 3, 4, 5, 6], f(x) = [0, 1, 5, 14, 30, 55].

Solution: The table corresponds to f(n) = ((n-1)n(2n-1) / 6) with step h = 1.
Differences at x_0 = 1:
Δ f_0 = 1 - 0 = 1, Δ^2 f_0 = 4 - 1 = 3, Δ^3 f_0 = 5 - 4 = 1.
Here x = 1.2 \implies u = (1.2 - 1 / 1) = 0.2.
Newton's forward difference derivative formula:
f'(x) = (1 / h) [Δ f_0 + (2u - 1 / 2)Δ^2 f_0 + (3u^2 - 6u + 2 / 6)Δ^3 f_0]
Substitute u = 0.2: f'(1.2) = 1 + (2(0.2) - 1 / 2)(3) + (3(0.04) - 6(0.2) + 2 / 6)(1)
= 1 + (-0.6 / 2)(3) + (0.12 - 1.2 + 2 / 6) = 1 - 0.90 + (0.92 / 6) = 0.10 + 0.1533 = **0.2533**.

Q32 **Marks: 3M** CCSU Exam **CCSU: 2020**

Define Simpson's three-eighths rule for Numerical Integration.

Definition: Simpson's 3/8 rule approximates the integral of f(x) over [a, b] divided into n subintervals where n is a multiple of 3 by fitting cubic polynomials across triplets of intervals:
∫_a^b f(x) \, dx ≈ (3h / 8) [(y_0 + y_n) + 3∑_{i ≠q 3k} y_i + 2∑_{i = 3k} y_i]
The truncation error is of order O(h^4), specifically E = -(3 / 80) h^5 f^{(4)}(\xi).

Q33 **Marks: 3M** CCSU Exam **CCSU: 2018**

Find the first derivative of f(x) at x = 0.4 from the table: x = [0.1, 0.2, 0.3, 0.4], y = [1.1051, 1.2214, 1.3498, 1.4918].

Solution: Since x = 0.4 is the last point in the equidistant table (h = 0.1), we use **Newton's Backward Difference formula** for derivative:
f'(x_n) = (1 / h) [∇ y_n + (1 / 2)∇^2 y_n + (1 / 3)∇^3 y_n]
Backward differences at x_n = 0.4:
∇y_3 = 1.4918 - 1.3498 = 0.1420
∇y_2 = 1.3498 - 1.2214 = 0.1284 $\implies \nabla^2 y_3 = 0.1420 - 0.1284 = 0.0136$
∇y_1 = 1.2214 - 1.1051 = 0.1163 $\implies \nabla^2 y_2 = 0.1284 - 0.1163 = 0.0121 \implies \nabla^3 y_3 = 0.0136 - 0.0121 = 0.0015$
Substitute into formula:
f'(0.4) = (1 / 0.1) [0.1420 + (1 / 2)(0.0136) + (1 / 3)(0.0015)] = 10 × [0.1420 + 0.0068 + 0.0005] = 10 × 0.1493 = **1.4930**

Verification: Since y ≈ e^x, e^{0.4} = 1.4918, exactly matching theory.